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Liu et al.

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- (54) **SEMICONDUCTOR STRUCTURE WITH ISOLATION STRUCTURE**

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H10D 84/80 (2025.01)
H10F 39/18 (2025.01)

- (52) **U.S. Cl.**
CPC ***H10F 39/807*** (2025.01); ***H10F 39/014***
(2025.01); ***H10F 39/18*** (2025.01); ***H10F***
39/8067 (2025.01)

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H10F 39/8067; H10F 39/18; H10F

39/8053; H10F 39/8063; H10F 19/75;
H01L 27/1463; H01L 21/76224-76237;
H01L 21/76229; H01L 21/76831; H01L
21/3065; H01L 21/31111; H01L
21/31116; H01L 21/762-76237; H10D
84/0151; H10D 84/0188; H10D 84/80;
H10D 30/611

See application file for complete search history.

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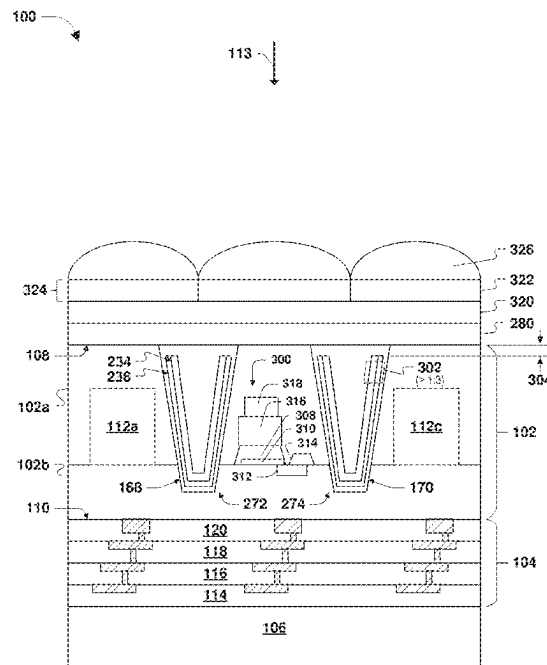
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(57) **ABSTRACT**

A semiconductor structure includes a first shallow trench isolation (STI) structure within a semiconductor substrate. The first STI structure includes a buffer structure, an adhesion structure, an electromagnetic reflection structure, and a fill structure. The adhesion structure is between and adhesively bonded to the buffer structure and the electromagnetic reflection structure. The electromagnetic reflection structure is between the adhesion structure and the fill structure to reflect electromagnetic radiation.

20 Claims, 23 Drawing Sheets



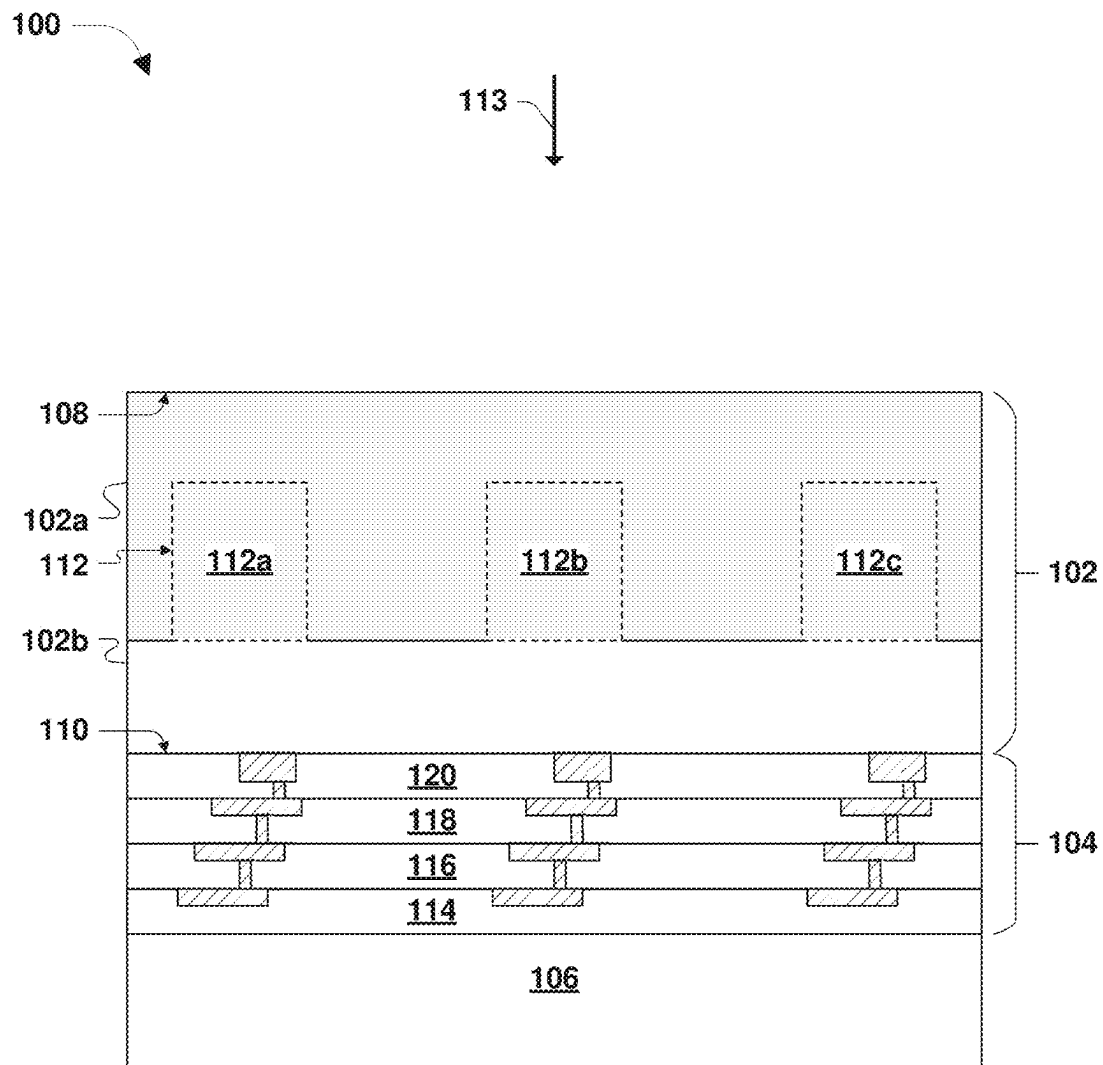


FIG. 1A

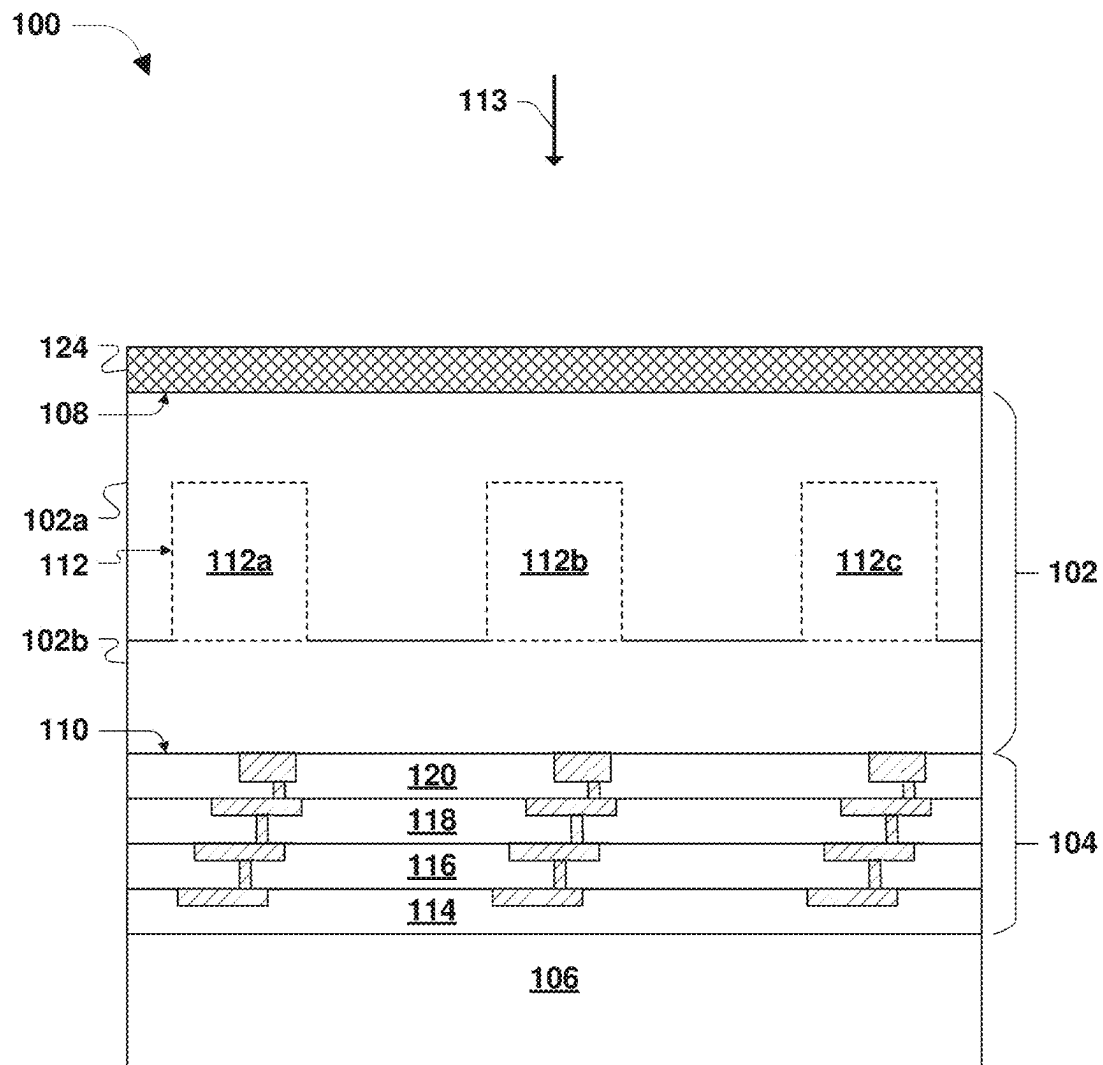


FIG. 1B

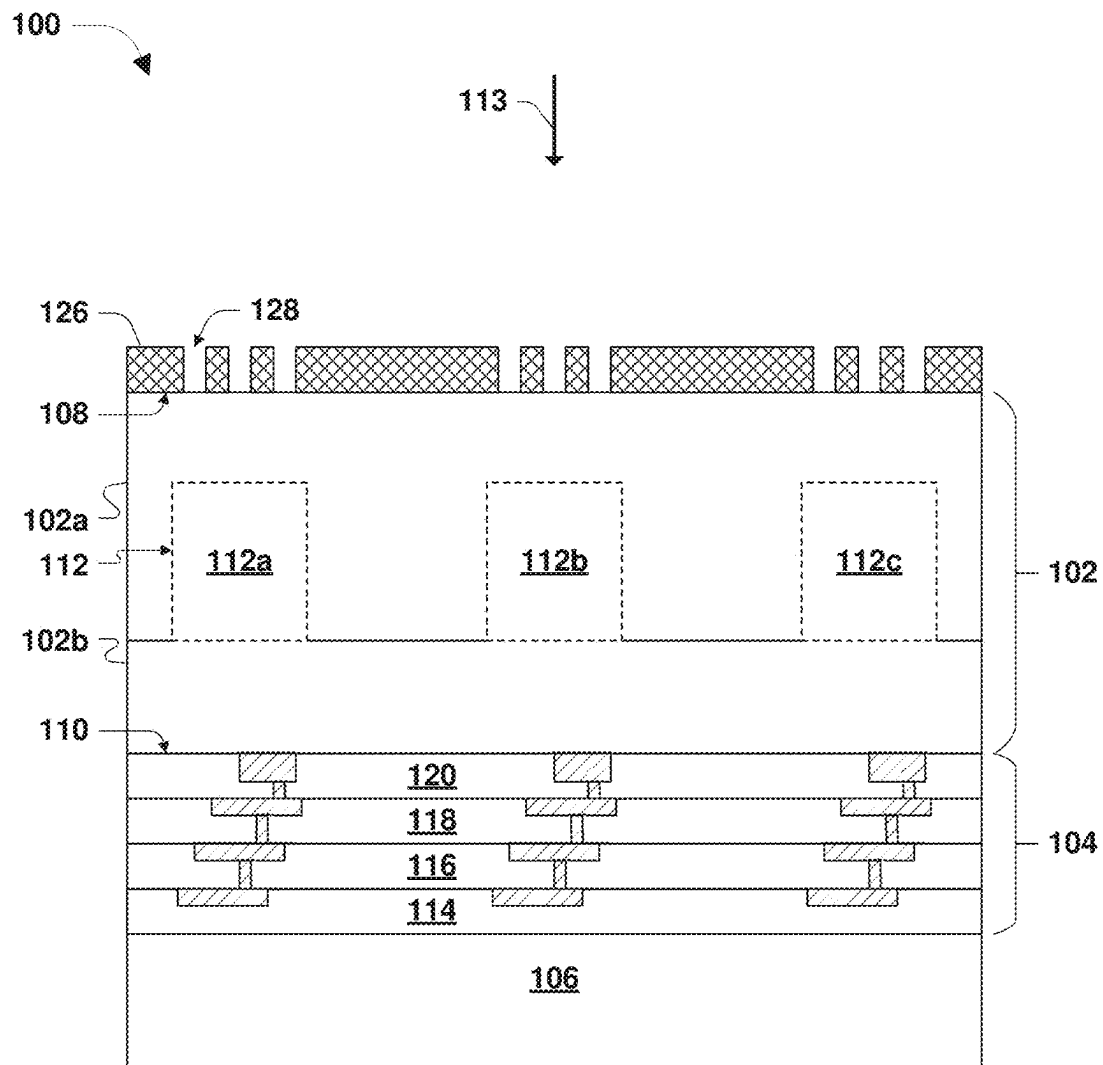


FIG. 1C

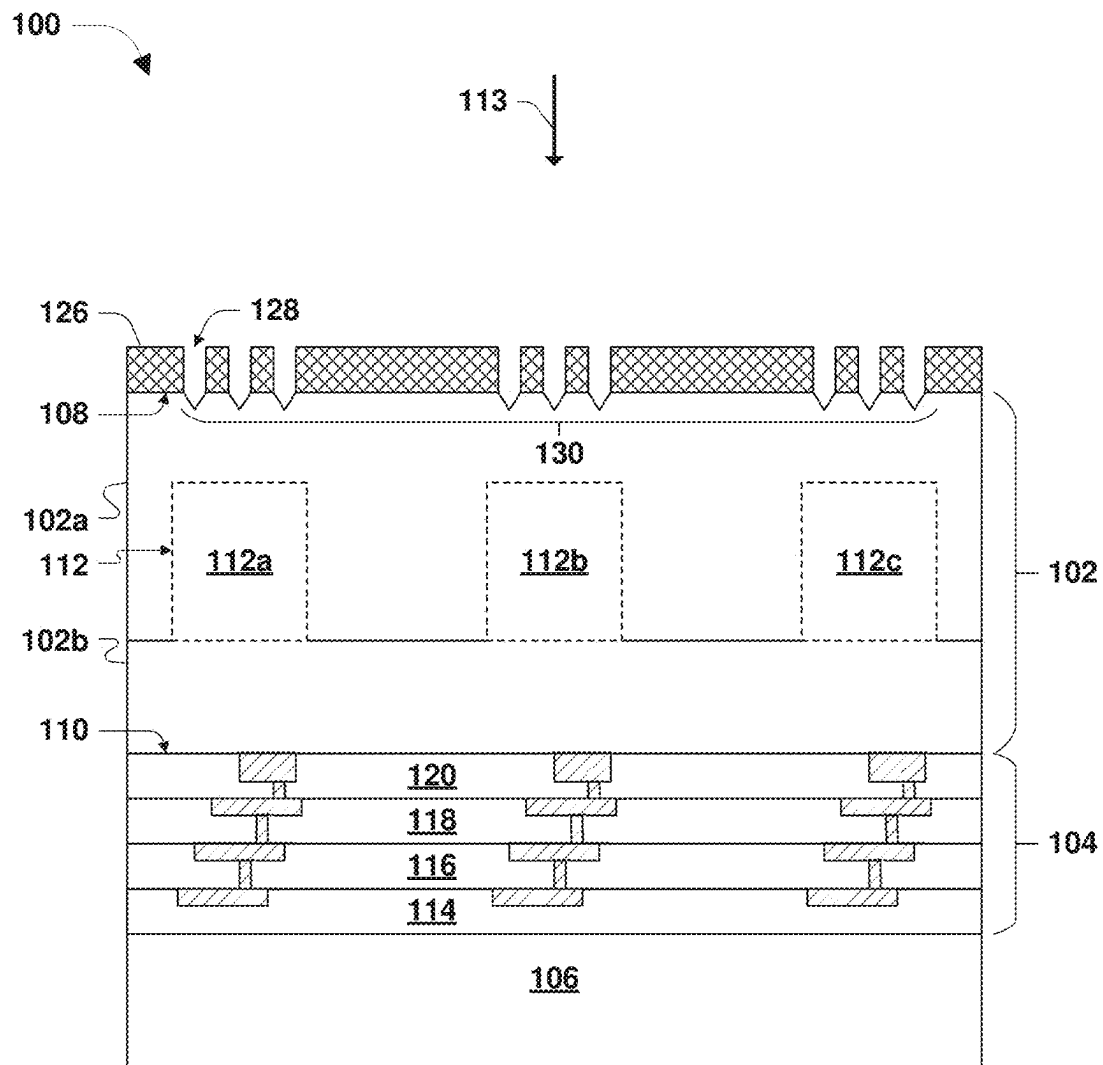


FIG. 1D

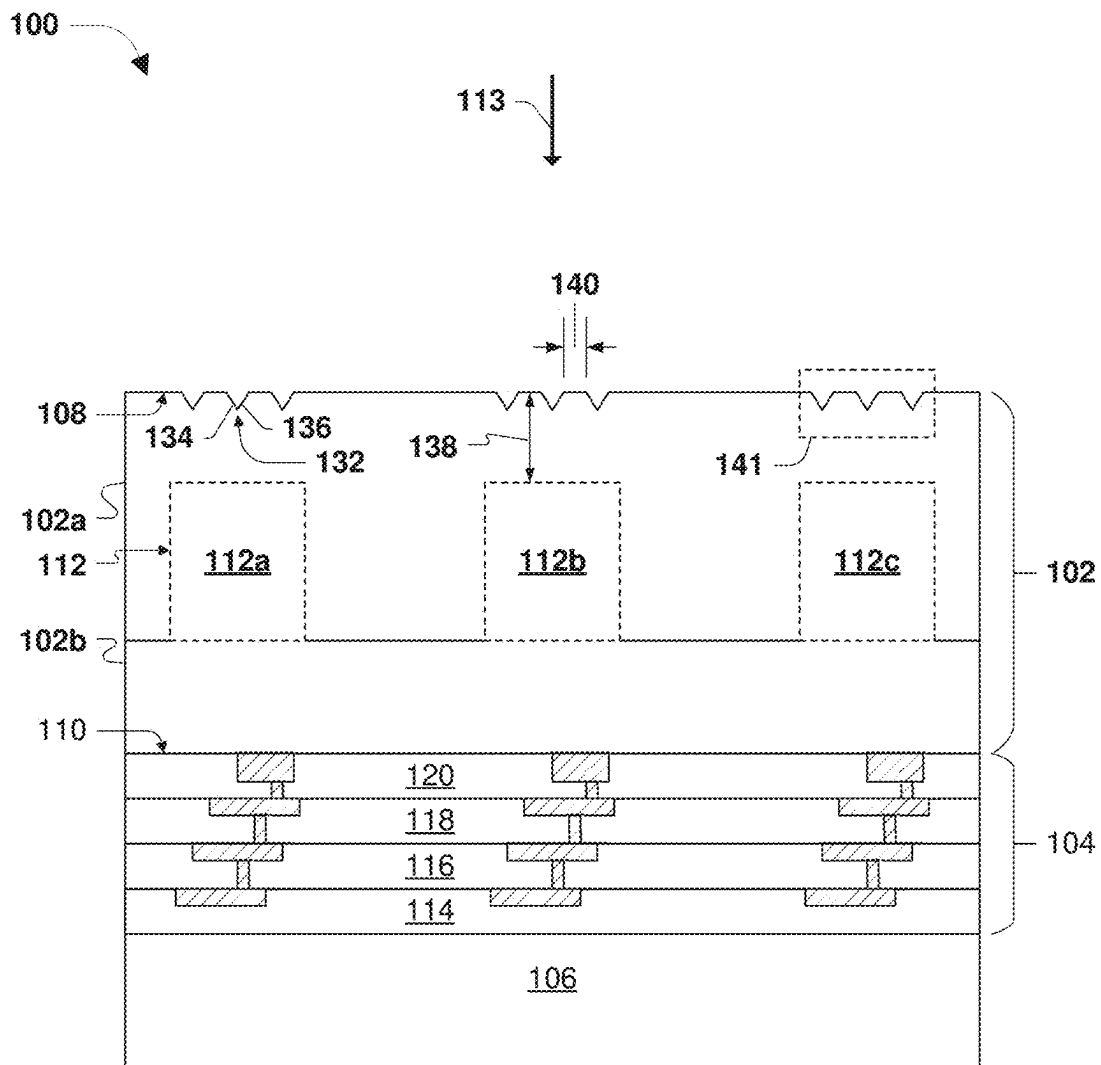


FIG. 1E

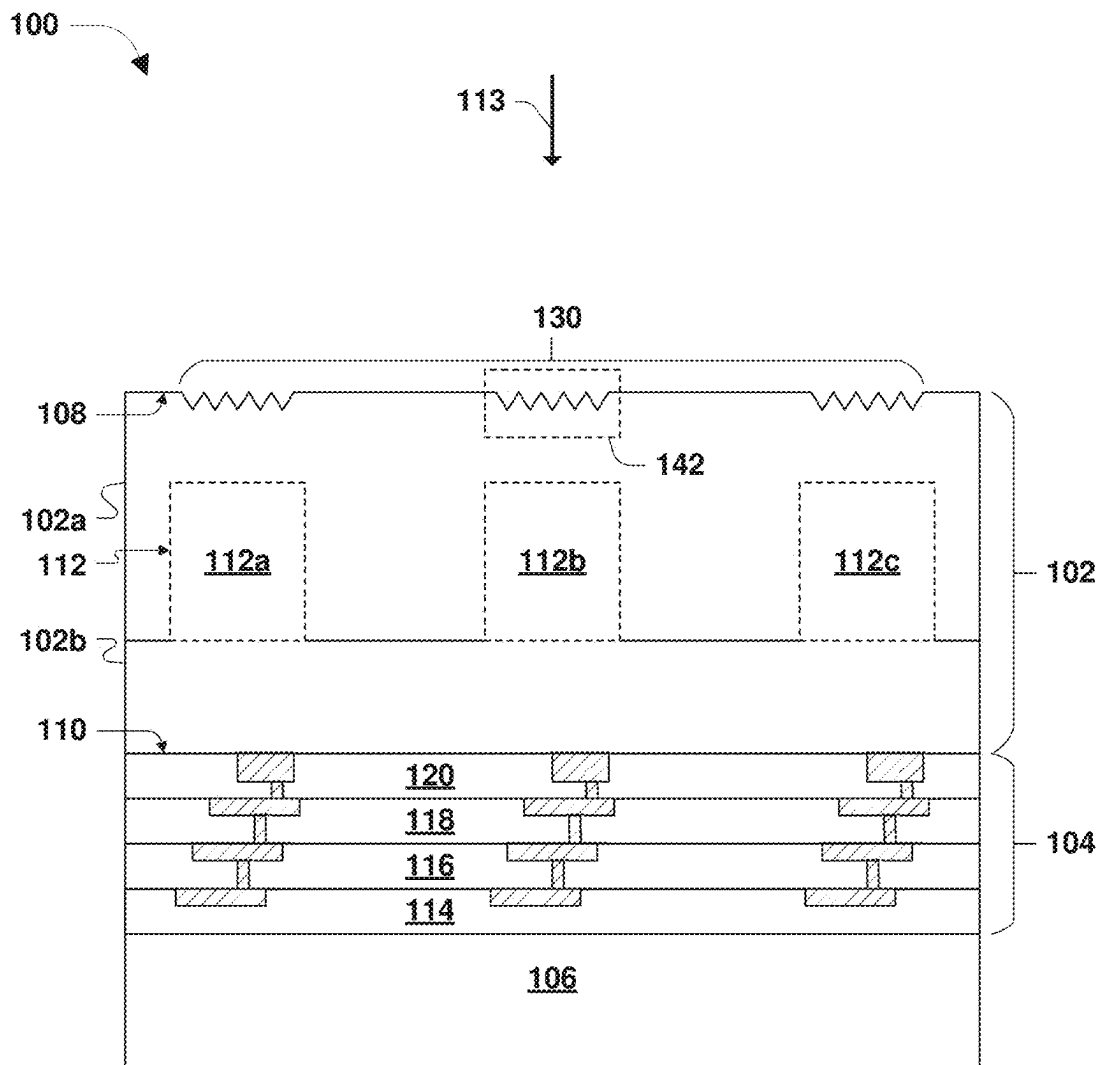


FIG. 1F

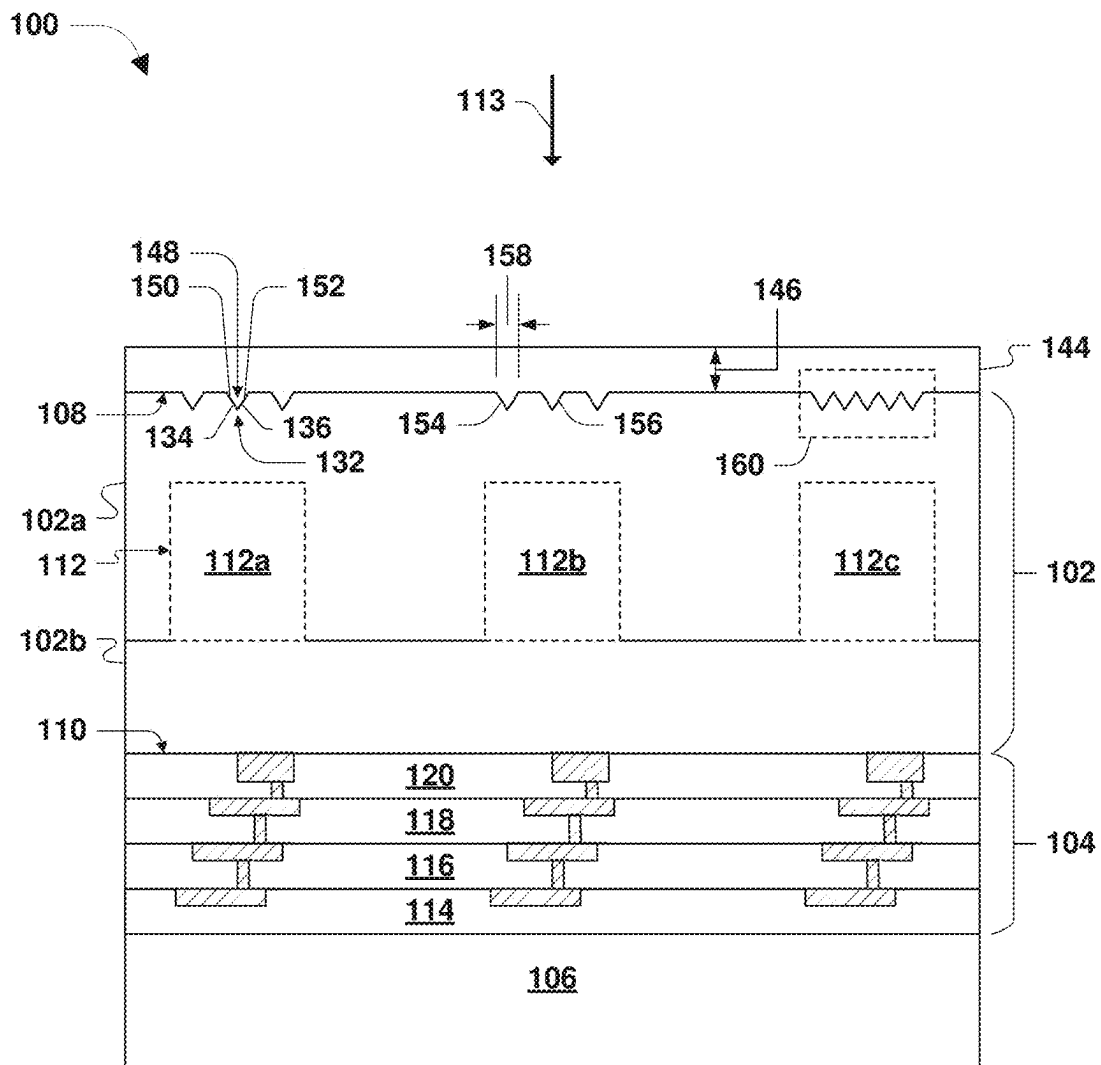


FIG. 1G

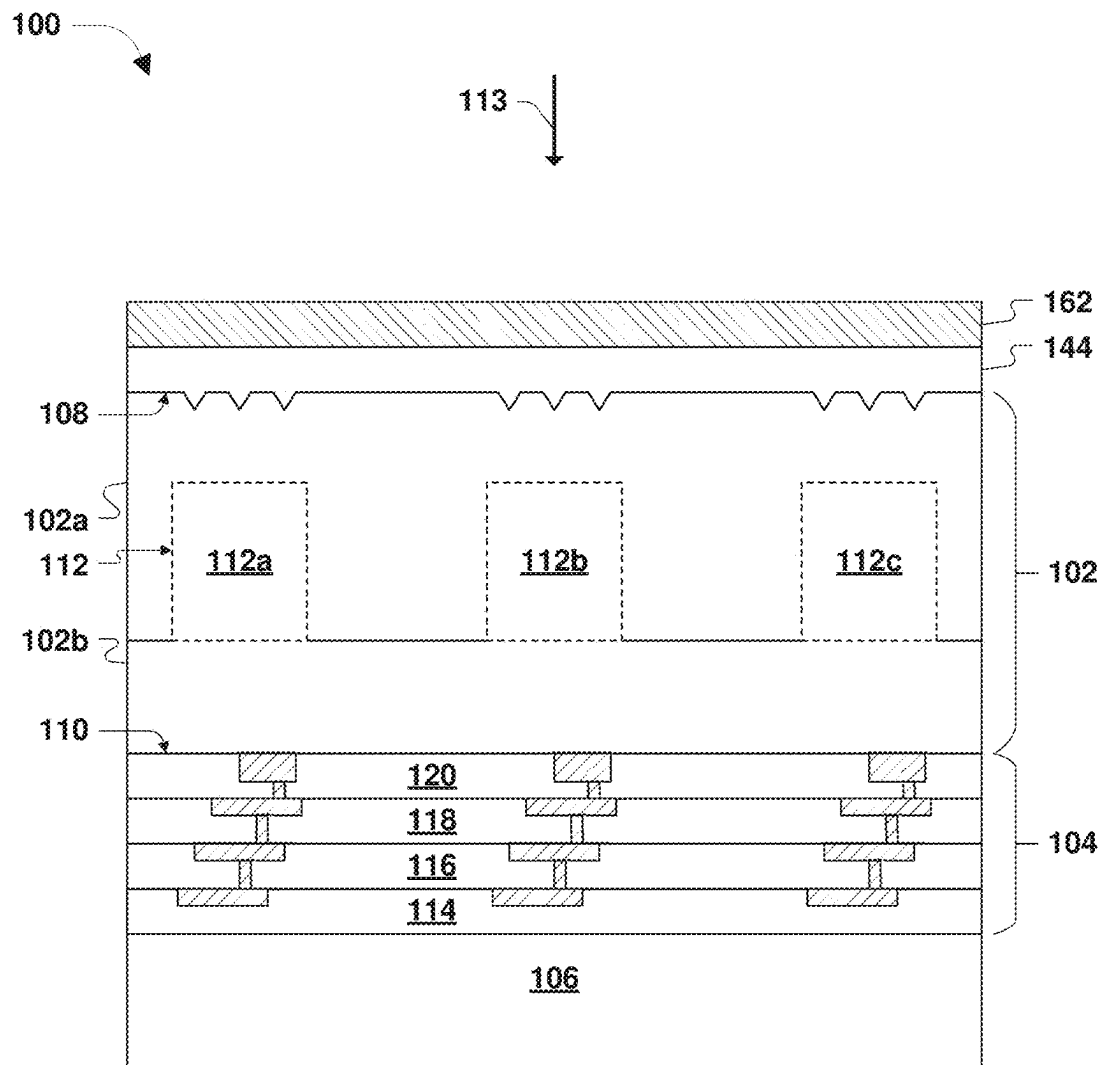


FIG. 1H

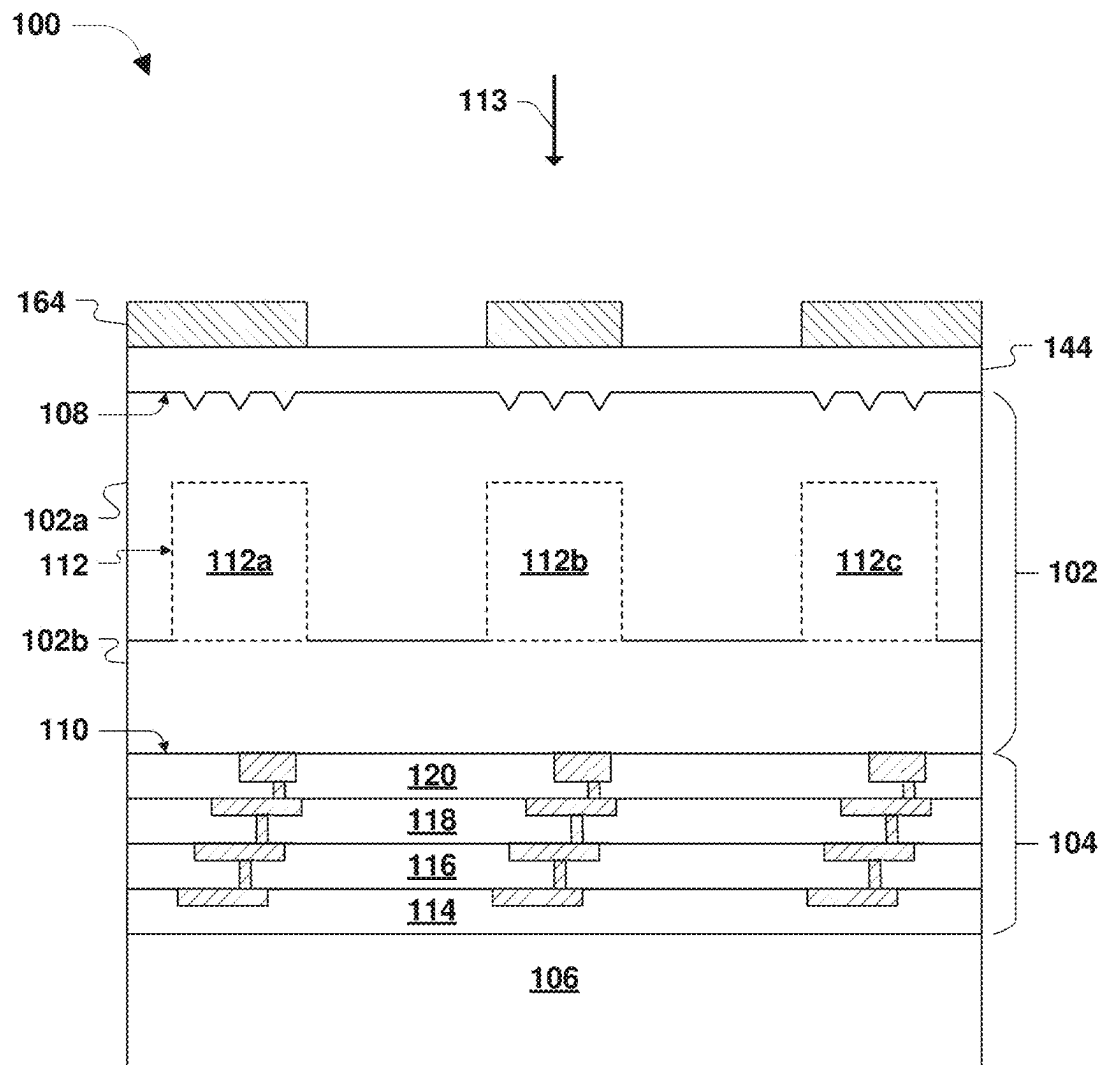


FIG. 11

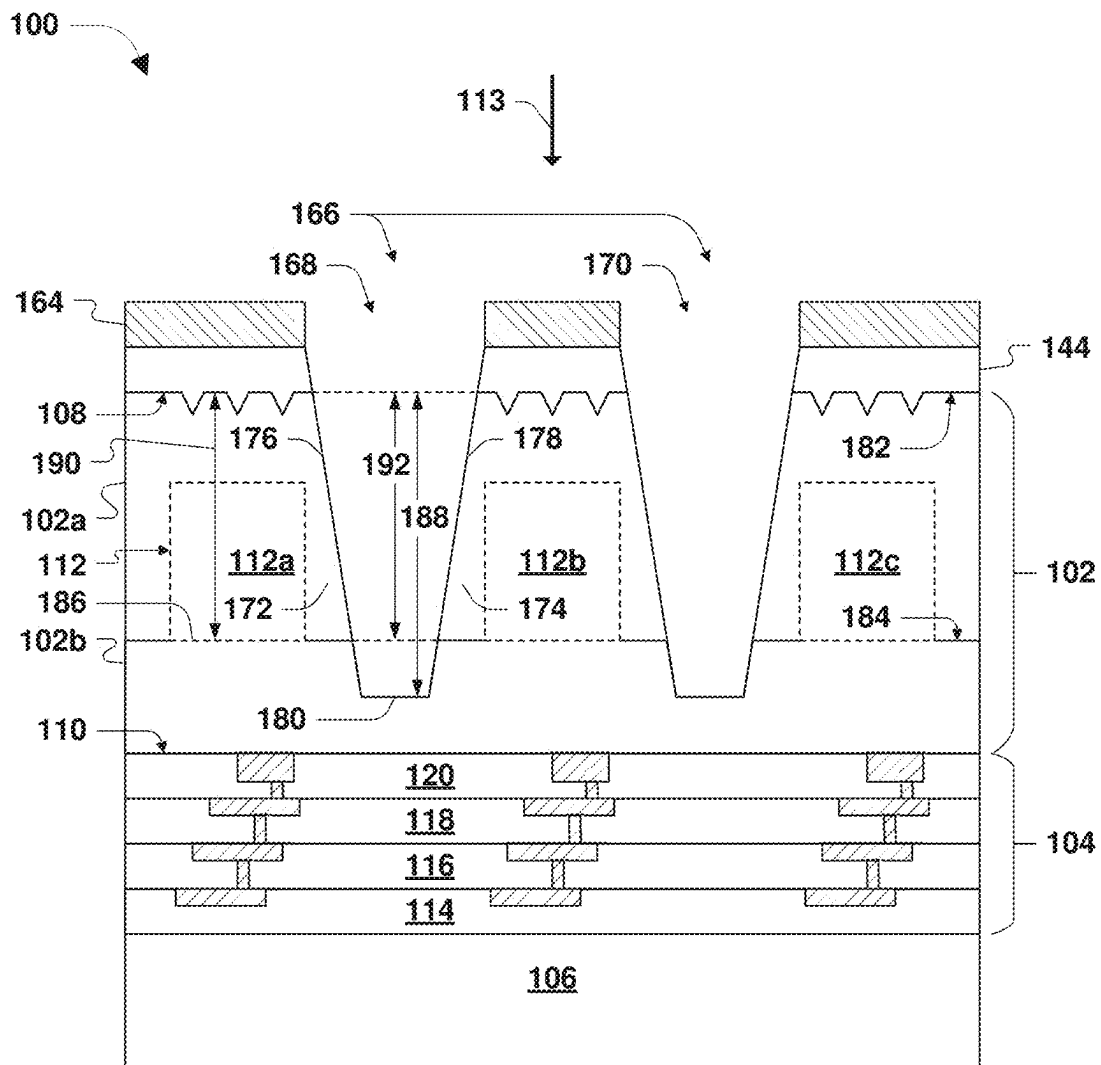


FIG. 1J

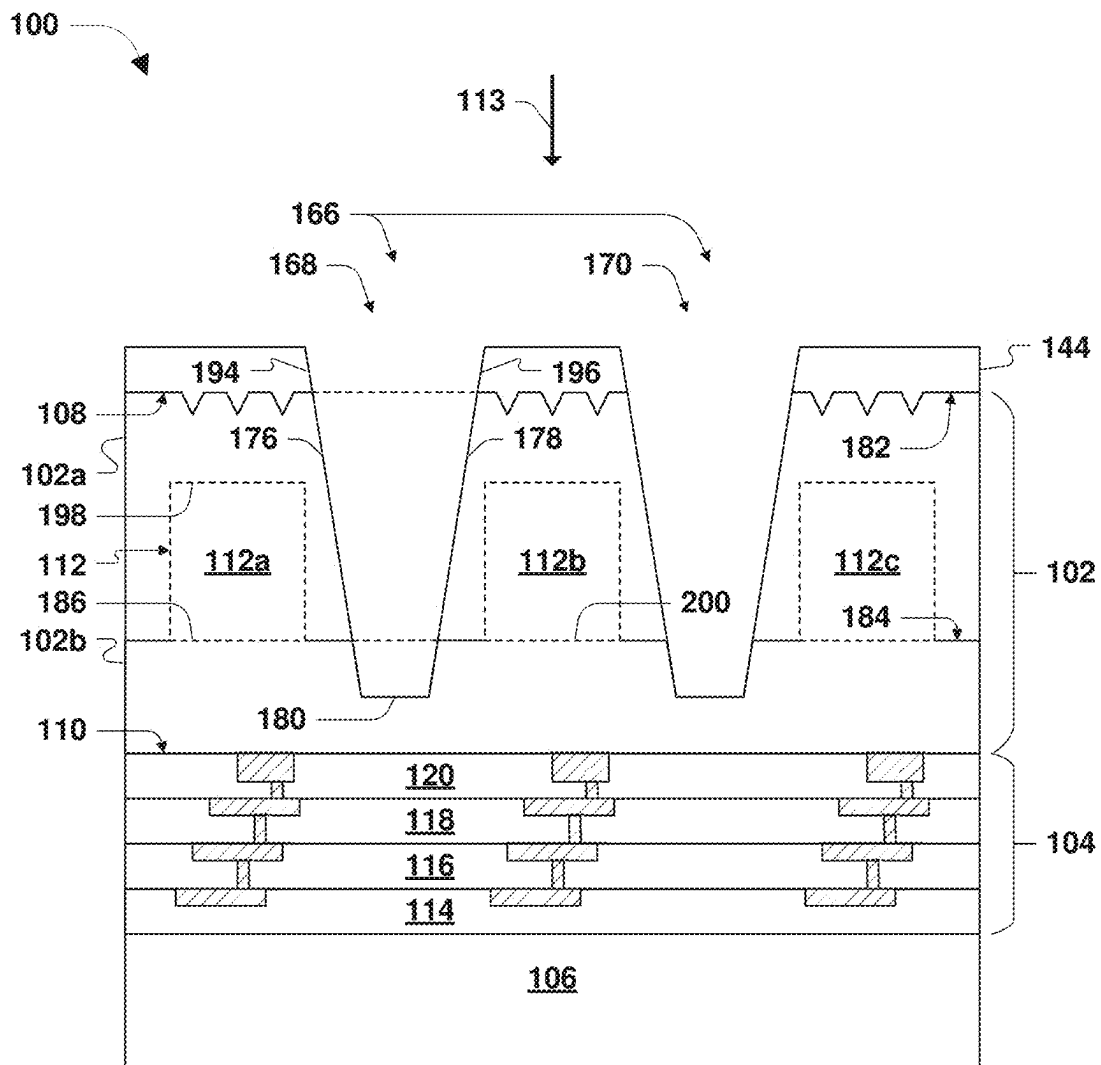


FIG. 1K

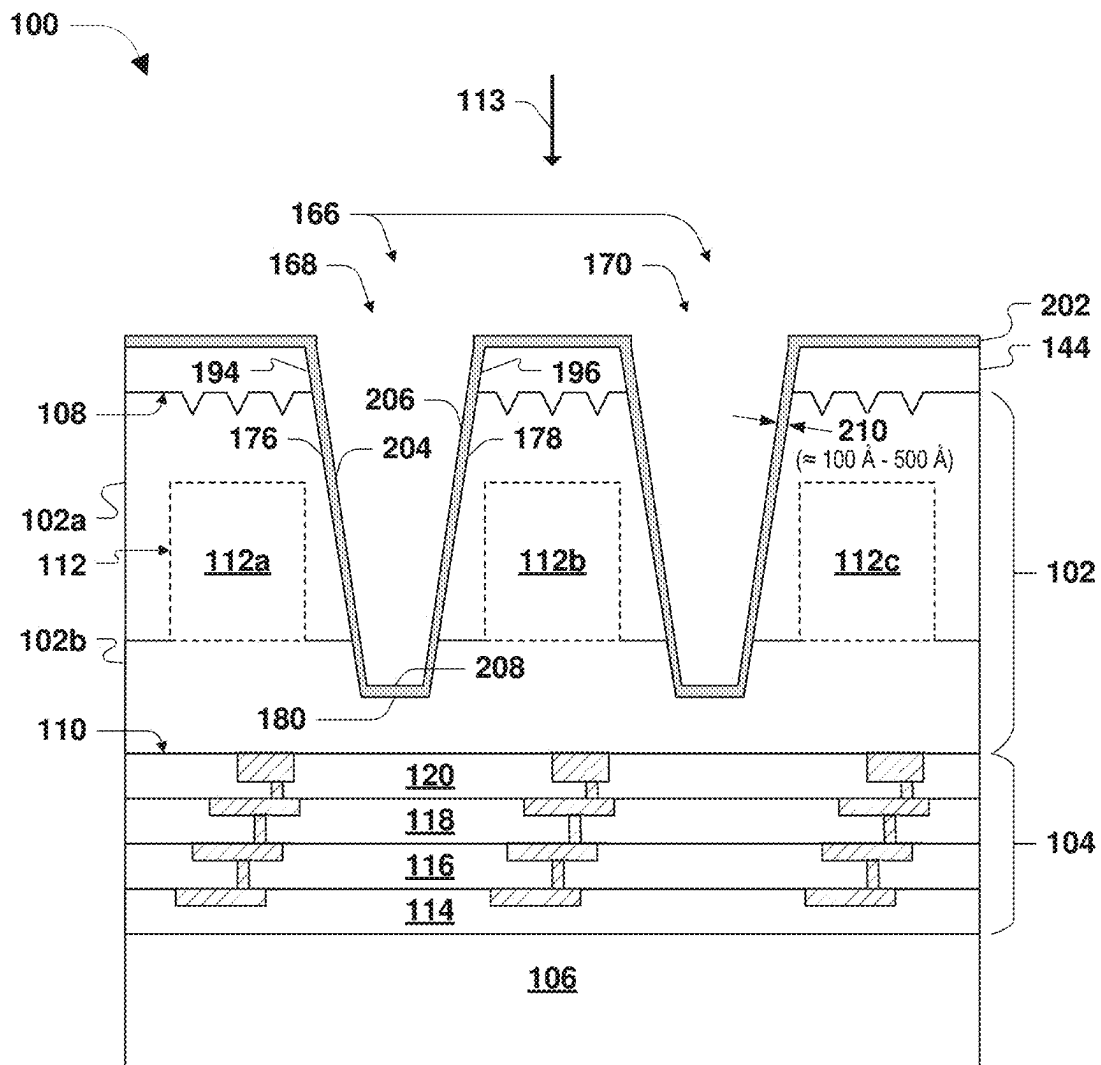


FIG. 1L

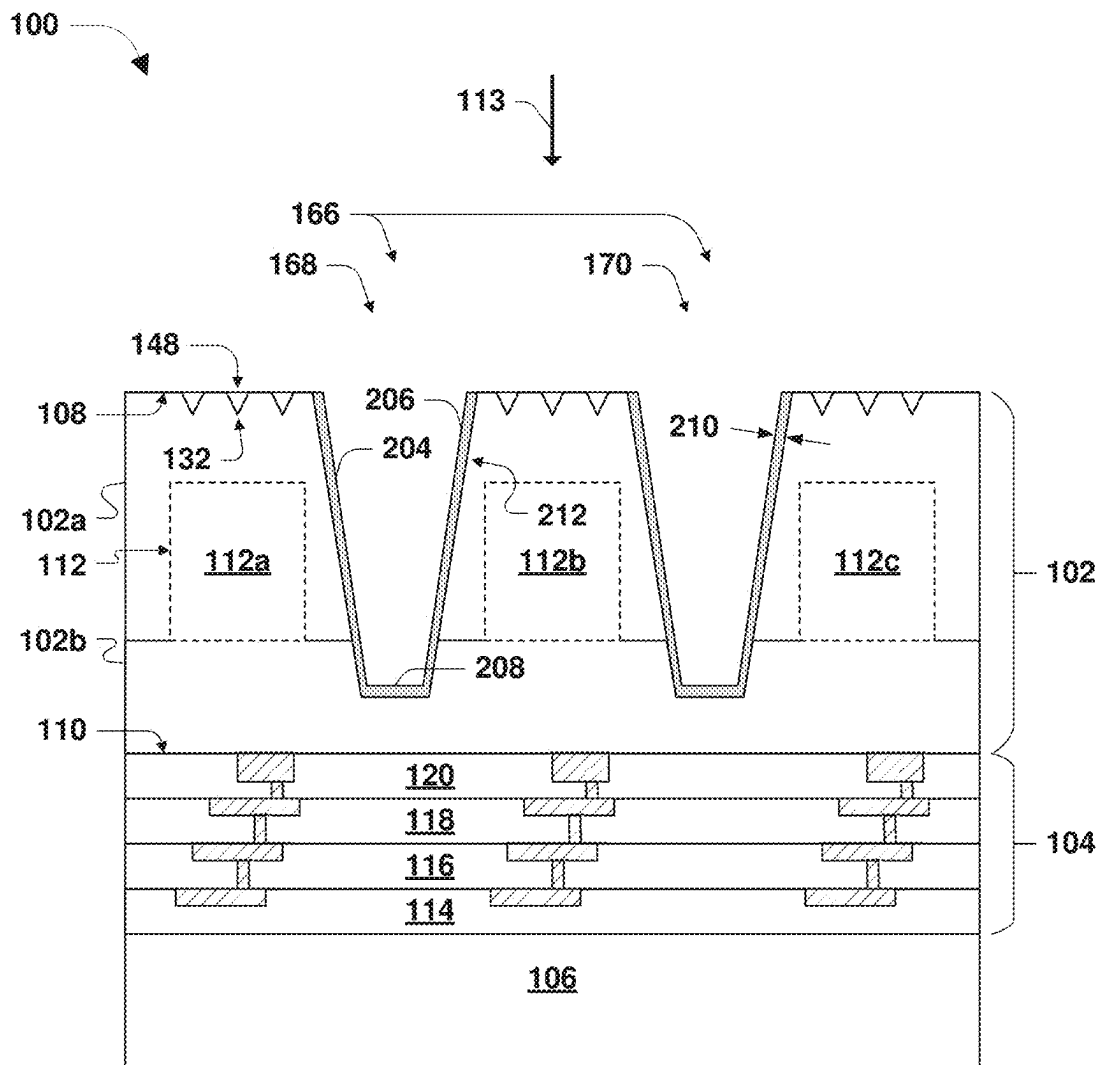


FIG. 1M

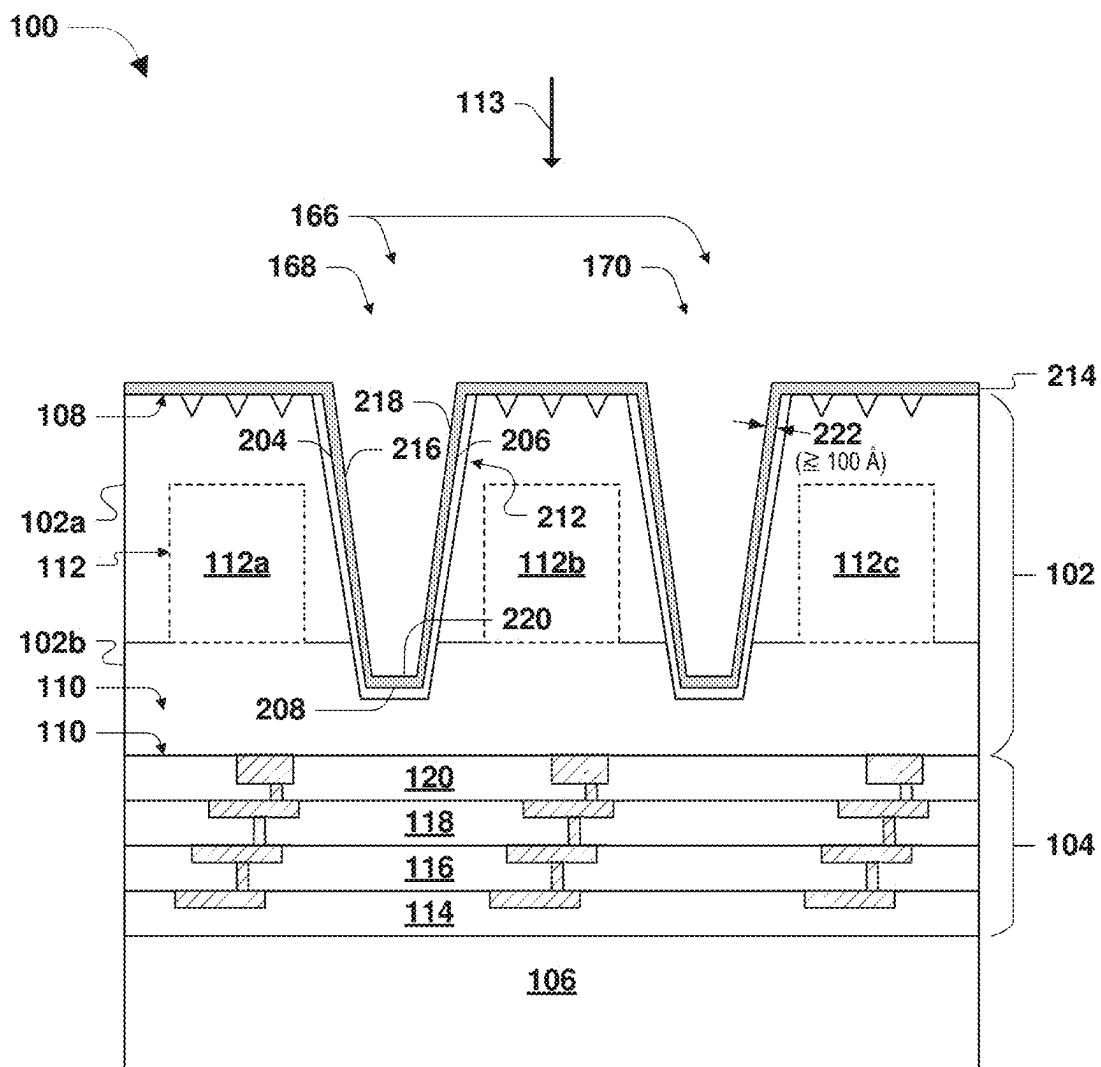


FIG. 1N

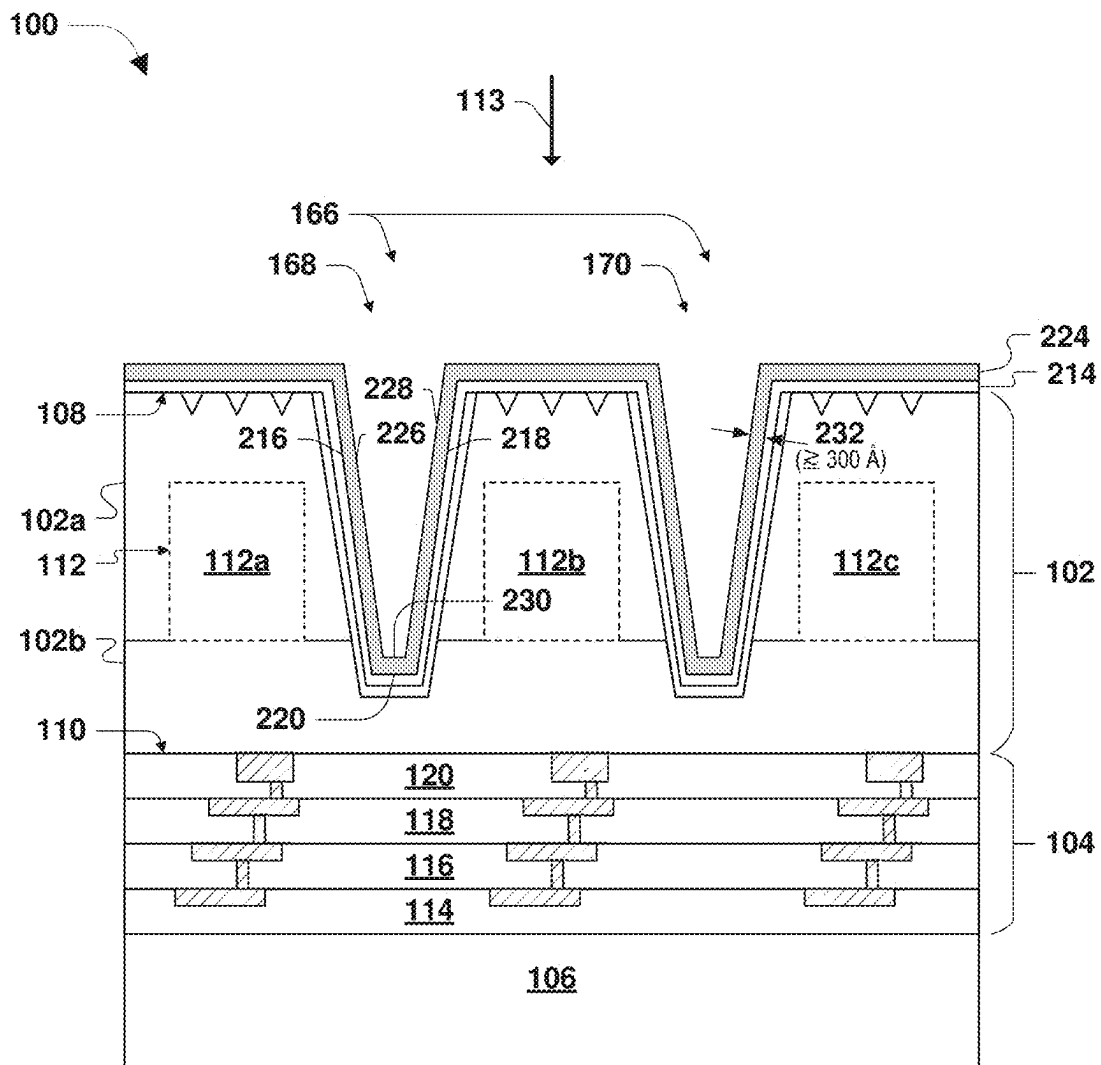


FIG. 10

FIG. 1P

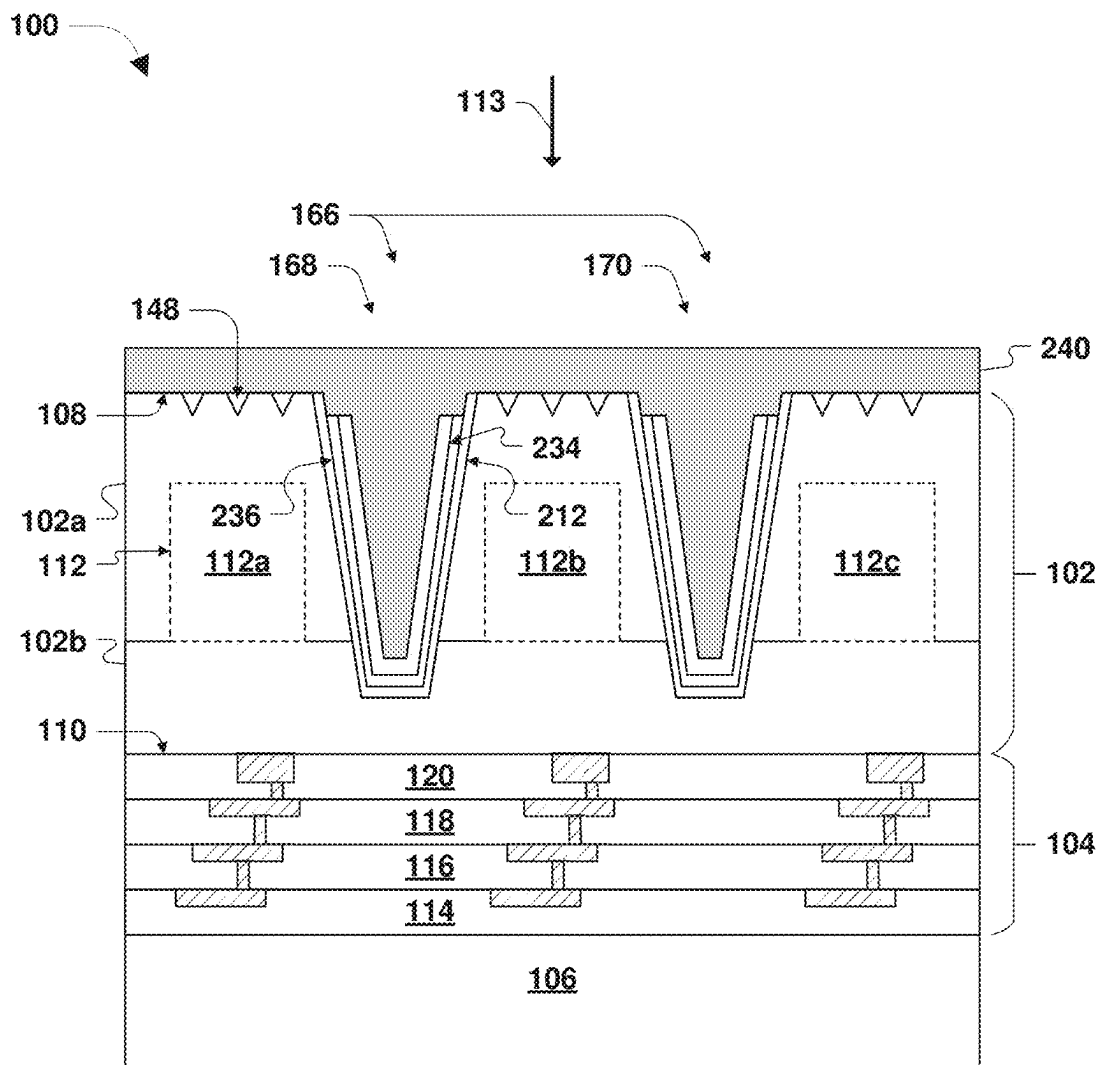


FIG. 1Q

FIG. 1R

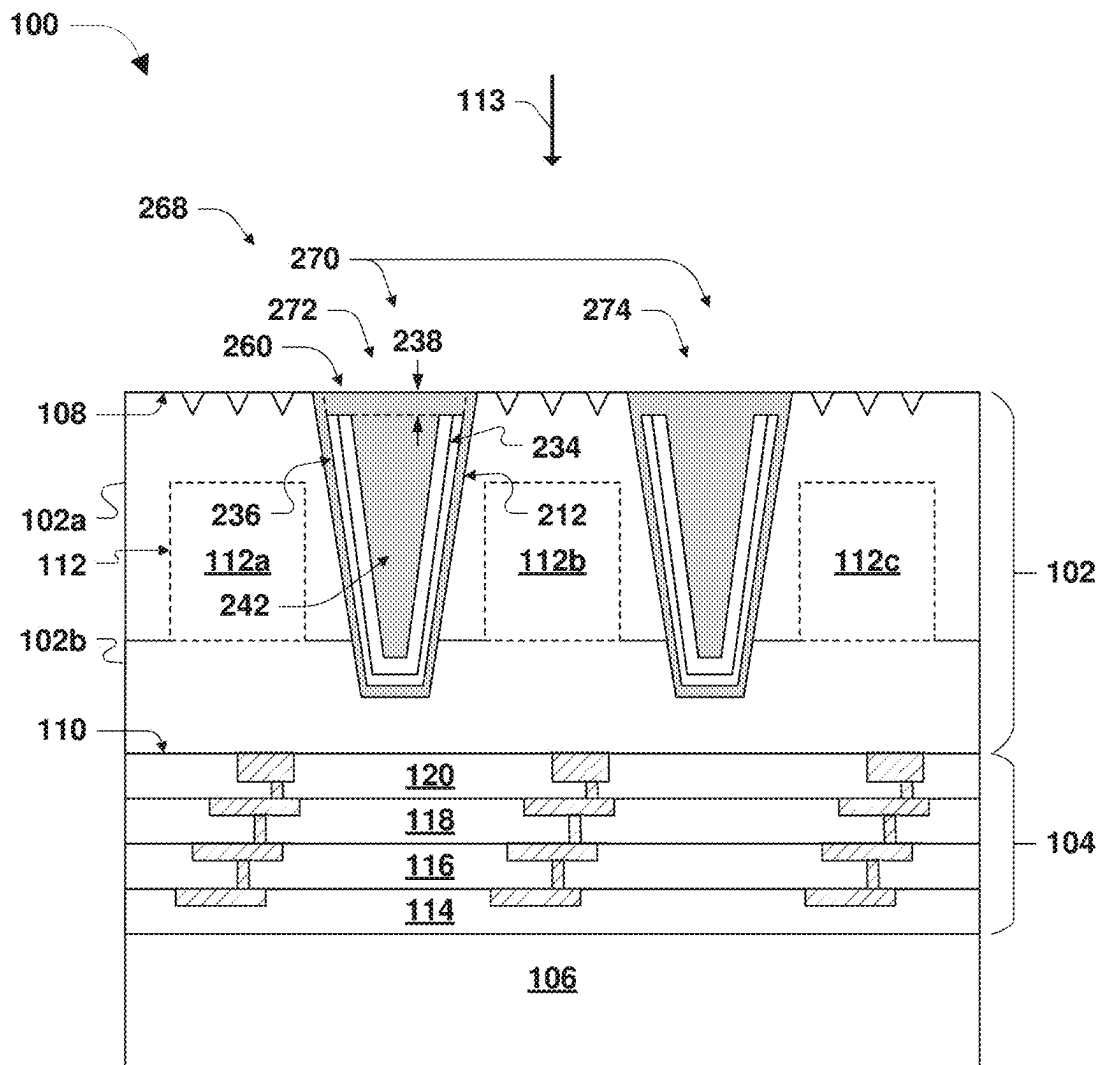


FIG. 1S

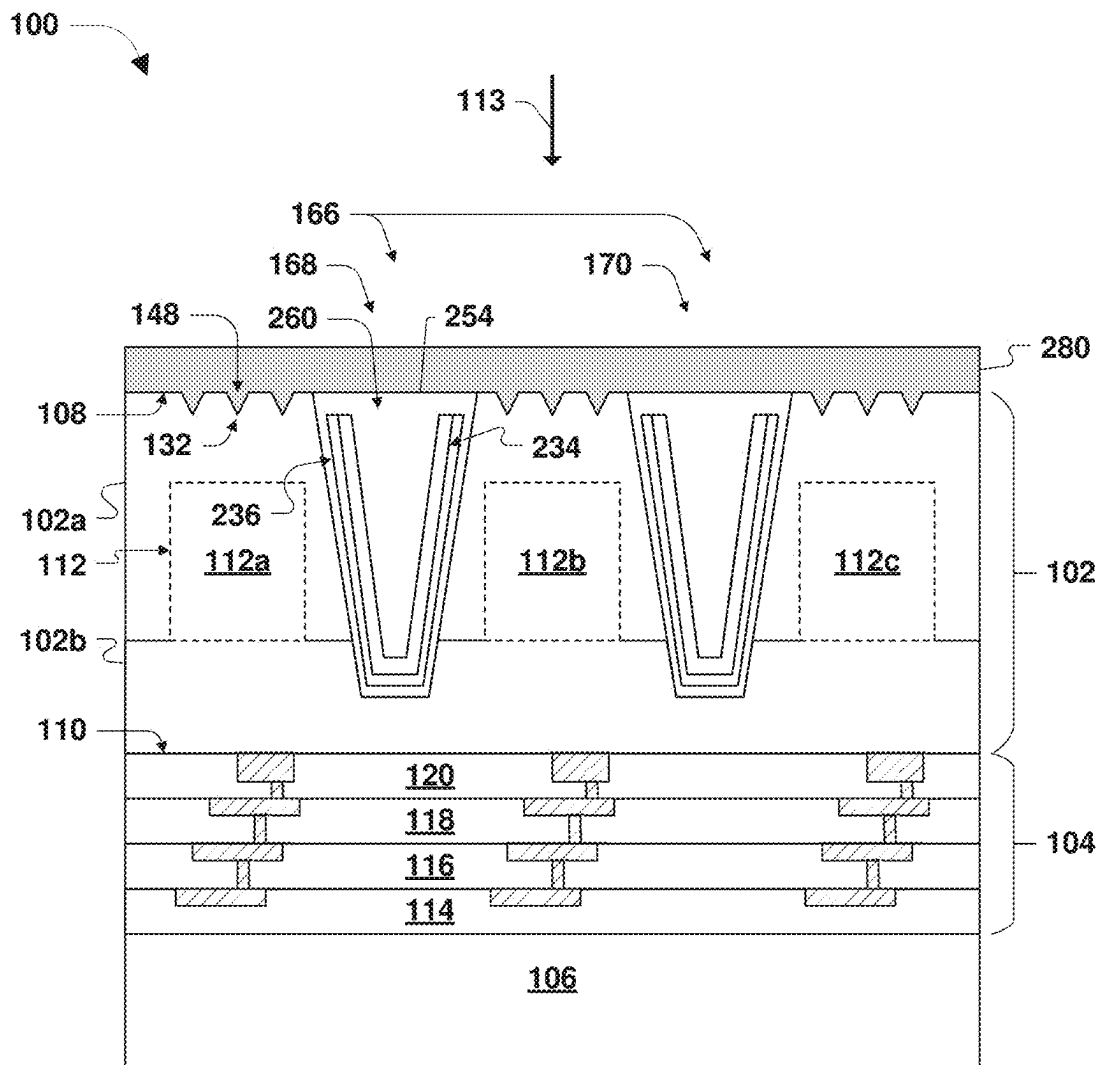


FIG. 1T

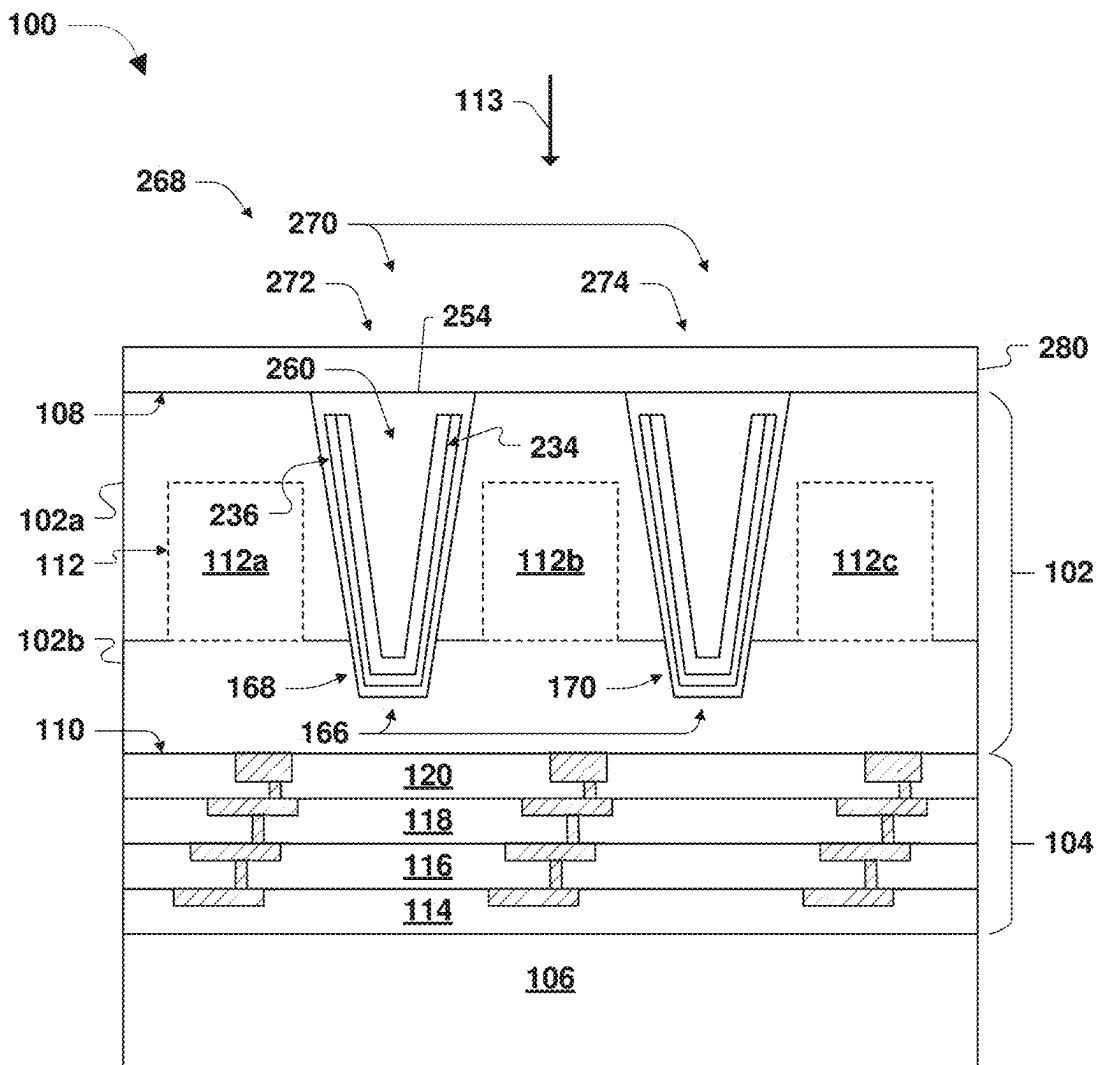


FIG. 2

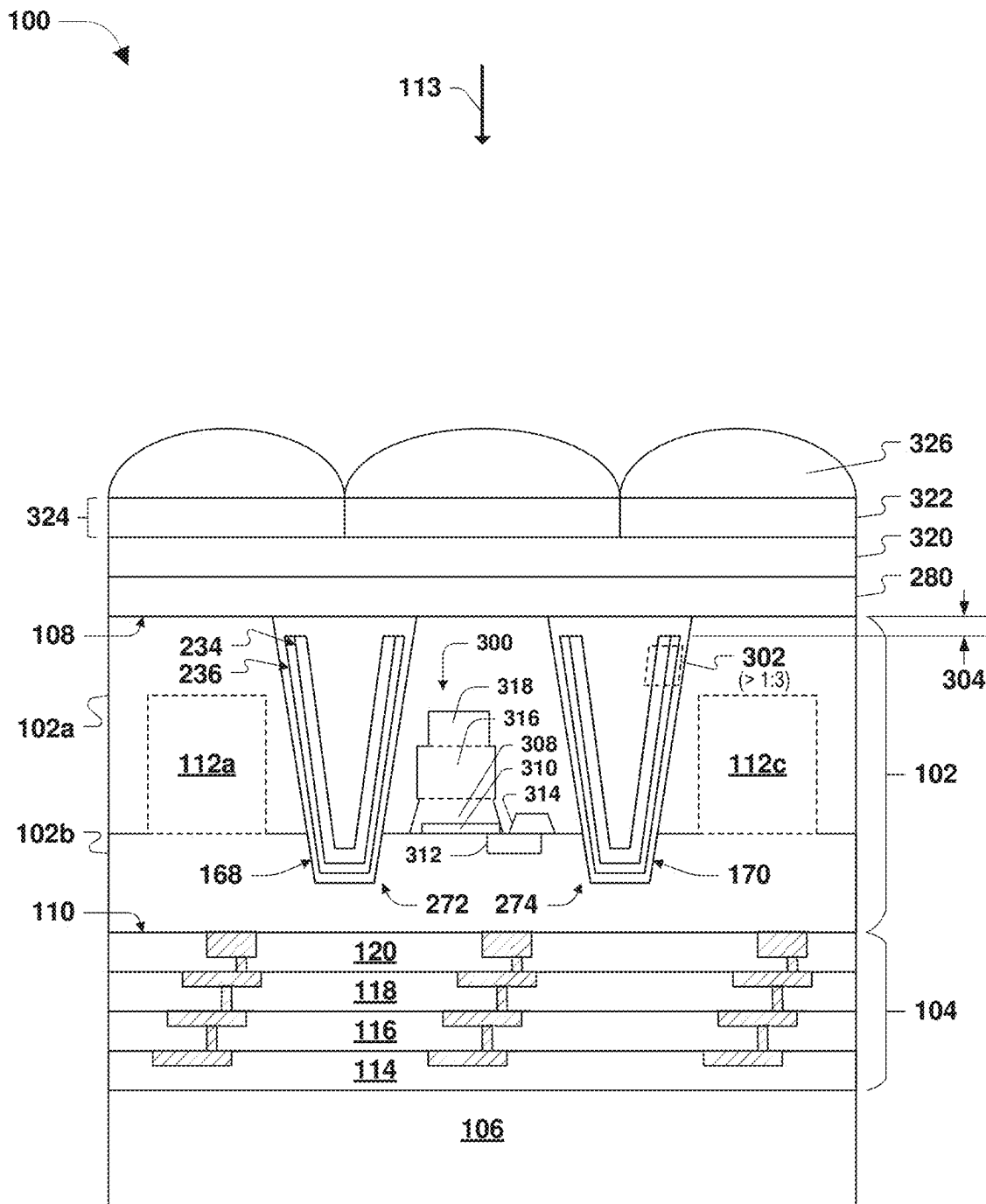
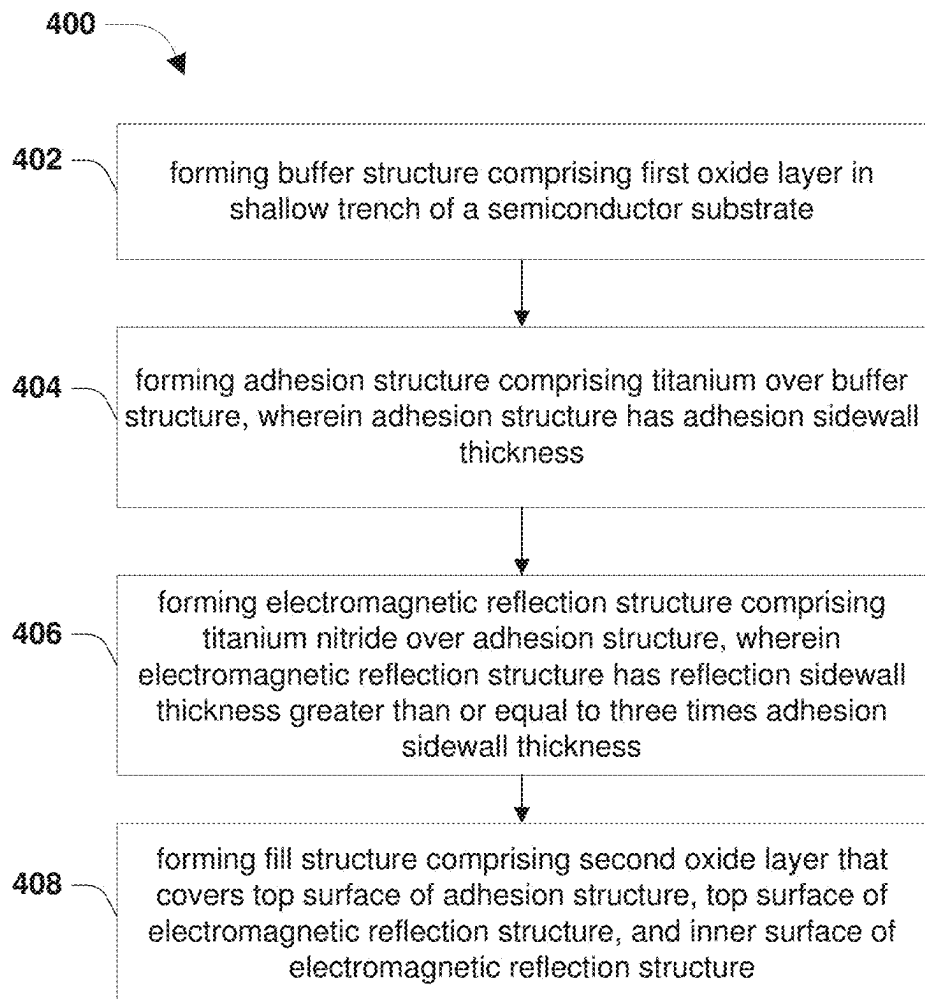


FIG. 3

**FIG. 4**

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SEMICONDUCTOR STRUCTURE WITH ISOLATION STRUCTURE

BACKGROUND

Semiconductor devices are used in a multitude of electronic devices, such as mobile phones, laptops, desktops, tablets, watches, gaming systems, and various other industrial, commercial, and consumer electronics. Semiconductor devices may be formed in a semiconductor substrate and may be isolated from other semiconductor devices by an isolation portion. Semiconductor devices may include a charge-coupled device (CCD), complementary metal-oxide-semiconductor (CMOS) radiation detecting elements, and/or other types of radiation detecting elements that are used to convert an image focused on a radiation detecting element into an electrical signal. The semiconductor device may comprise an array of radiation detecting elements, such as photodiodes, configured to produce an electrical signal corresponding to an intensity of radiation impinging on the radiation detecting element. The electrical signal is used to display a corresponding image on a monitor or provide information about the optical image.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A-1T are illustrations of a semiconductor arrangement at various stages of fabrication, according to some embodiments.

FIG. 2 is a cross-sectional view of a semiconductor arrangement, according to some embodiments.

FIG. 3 is a cross-sectional view of a semiconductor arrangement 100, according to some embodiments.

FIG. 4 illustrates an example method, according to some embodiments.

DETAILED DESCRIPTION

The following disclosure provides several different embodiments, or examples, for implementing different features of the provided subject matter with reference to the drawings, wherein like reference numerals are generally used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide an understanding of the claimed subject matter. It is evident, however, that the claimed subject matter can be practiced without these specific details. In other instances, structures and devices are illustrated in block diagram form in order to facilitate describing the claimed subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may

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repeat reference numerals or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation illustrated in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Also, relationship terms such as “connected to,” “adjacent to,” “coupled to,” and the like, may be used herein to describe both direct and indirect relationships. “Directly” connected, adjacent, or coupled may refer to a relationship in which there are no intervening components, devices, or structures. “Indirectly” connected, adjacent, or coupled may refer to a relationship in which there are intervening components, devices, or structures.

One or more semiconductor structures and one or more techniques for forming such semiconductor structures are provided herein. According to some embodiments, a semiconductor structure and method of forming a semiconductor structure are provided. In an example, a semiconductor structure comprises a first shallow trench isolation (STI) structure within a semiconductor substrate. The first STI structure may be laterally offset from a semiconductor device, such as a CMOS radiation detecting element, a photodiode, or other type of semiconductor device. The first STI structure may provide a barrier to electromagnetic radiation. For example, the first STI structure may reflect electromagnetic radiation toward the semiconductor device, such as a photodiode. The first STI structure may provide electrical isolation between semiconductor devices, such as a first photodiode and a second photodiode.

According to some embodiments, a semiconductor arrangement comprises a photodiode array formed within a substrate and an STI structure array disposed between and laterally offset from the photodiode array. The photodiode array may comprise one or more photodiodes, such as image sensor pixels, configured to accumulate energy generated by optical radiation, near-infrared (NIR) radiation, or other type of electromagnetic radiation, such as from photons, of an optical image. A voltage of a photodiode can be read as an output for the optical image. In some embodiments, a photodiode is situated under one or more layers or components formed over a substrate. Because radiation travels along a path that comprises such layers or components before reaching the photodiode, signal strength of the radiation can decay before reaching the photodiode or the radiation can travel towards another photodiode. For example, the radiation can be detected by a neighboring or adjacent photodiode, which can result in crosstalk. Crosstalk can degrade performance of the semiconductor arrangement, increase noise, and/or decrease at least one of quality or intensity of signals produced by the semiconductor arrangement. STI structures, as set forth in greater detail herein, may reduce and/or eliminate crosstalk between such photodiodes.

In some embodiments, STI structures, as set forth herein, may improve quantum efficiency (QE) of semiconductor devices, such as photodiodes. In some embodiments, QE refer to a ratio of an incident photon to a converted electron (IPCE) in a photosensitive device, such as a photodiode. For

example in a CCD or other photodetector, the QE may refer to a ratio between a number of charge carriers collected at a CCD terminal and a number of photons hitting a CCD photoreactive surface. As a ratio, QE may be dimensionless, but may be related to responsivity of the device, which may be expressed in amps per watt. Energy of a photon is inversely proportional to wavelength. Hence, QE may be measured over a range of different wavelengths to characterize device efficiency at a number of different photon energy levels. In some semiconductor photodiodes, QE may drop to zero for photons having an energy level below a band gap of such photodiode.

According to some embodiments, a semiconductor arrangement may include semiconductor devices such as photodiodes, pinned layer photodiodes, reset transistors, source follower transistors, floating diffusions (also known as floating diodes), transfer transistors, or other types of semiconductor devices. A CMOS semiconductor arrangement may include a CMOS active pixel image sensor (APS) with an intra-pixel charge transfer to a floating diffusion (FD). A pinned photodiode (PPD), also known as a pinned layer photodiode, is an example photodiode structure used in a CCD, a CMOS semiconductor arrangement, or a CMOS APS. A PPD provides, for example, at least one of low noise, high quantum efficiency, or low dark current. A CMOS semiconductor arrangement may be a front side illuminated (FSI) image sensor, detecting radiation from a front side, or a back side illuminated (BSI) image sensor, detecting radiation from a back side.

Accordingly, a semiconductor structure is provided herein. According to some embodiments, the semiconductor structure comprises a first shallow trench isolation (STI) structure within a semiconductor substrate, wherein the first STI structure comprises a buffer structure, an adhesion structure, an electromagnetic reflection structure, and a fill structure. The adhesion structure is between and adhesively bonded to the buffer structure and the electromagnetic reflection structure. The electromagnetic reflection structure is between the adhesion structure and the fill structure to reflect electromagnetic radiation. In some embodiments, the buffer structure comprises an oxide layer formed within a shallow trench of the semiconductor substrate. In some embodiments, the buffer structure is contiguous with a doped region of the semiconductor substrate. In some embodiments, the buffer structure has a buffer sidewall thickness between about 100 Angstroms (Å) to about 500 Å to buffer against dopants of the doped region. In some embodiments, the adhesion structure comprises titanium and is formed as an adhesion layer contiguous with the buffer structure. In some embodiments, the adhesion structure has an adhesion sidewall thickness greater than about 100 Å. In some embodiments, the electromagnetic reflection structure comprises titanium nitride. In some embodiments, the electromagnetic reflection structure is formed as a reflection layer contiguous with the adhesion structure. In some embodiments, the electromagnetic reflection structure has a reflection sidewall thickness greater than the adhesion sidewall thickness. In some embodiments, the reflection sidewall thickness is greater than or equal to three times the adhesion sidewall thickness.

In some embodiments, the buffer structure comprises a first oxide layer formed within a shallow trench of the semiconductor substrate. In some embodiments, the fill structure comprises a second oxide layer that covers a top surface of the adhesion structure, a top surface of the electromagnetic reflection structure, and an inner surface of the electromagnetic reflection structure. In some embodi-

ments, the second oxide layer has a second oxide top layer thickness greater than or equal to 100 Å over the top surface of the adhesion structure and the top surface of the electromagnetic reflection structure. In some embodiments, the first oxide layer has a greater refractive index than the second oxide layer. In some embodiments, the fill structure comprises at least one of silicon dioxide, silicon nitride, silicon oxynitride, hafnium oxide, or fluorinated silica glass. In some embodiments, the buffer structure comprises at least one of aluminum oxide, hafnium oxide, or tantalum nitride. In some embodiments, the adhesion structure comprises a mixture of titanium and a first metal. In some embodiments, the electromagnetic reflection structure comprises a mixture of titanium nitride and a second metal.

In some embodiments, the semiconductor structure comprises a semiconductor device within the semiconductor substrate, wherein the first STI structure is laterally offset from the semiconductor device. In some embodiments, the semiconductor structure comprises a second STI structure within the semiconductor substrate and laterally offset from the semiconductor device, wherein the semiconductor device is between the first STI structure and the second STI structure. In some embodiments, the first STI structure extends into the semiconductor substrate a first depth from a top side of the semiconductor substrate, and the semiconductor device extends into the substrate a second depth, less than the first depth, from the top side of the semiconductor substrate.

According to some embodiments, a semiconductor structure comprises a photodiode within a semiconductor substrate. A first shallow trench isolation (STI) structure is laterally offset from the photodiode within the semiconductor substrate. The first STI structure comprises a buffer structure including a first oxide layer and formed within a shallow trench of the semiconductor substrate. The first STI structure comprises an adhesion structure including titanium and formed as an adhesion layer over the first oxide layer. The first STI structure comprises an electromagnetic reflection structure including titanium nitride and formed as a reflection layer over the adhesion layer. The first STI structure comprises a fill structure including a second oxide layer that covers a top surface of the adhesion structure, a top surface of the electromagnetic reflection structure, and an inner surface of the electromagnetic reflection structure. In some embodiments, the shallow trench has a first trench tapered sidewall tapered toward the photodiode and the buffer structure has a first buffer tapered sidewall that aligns with the first trench tapered sidewall. In some embodiments, the adhesion structure has a first adhesion tapered sidewall that aligns with the first buffer tapered sidewall and the electromagnetic reflection structure has a first reflection tapered sidewall that aligns with the first adhesion tapered sidewall.

One or more methods of forming a semiconductor structure are provided herein. According to some embodiments, a buffer structure comprising a first oxide layer is formed in a shallow trench of a semiconductor substrate. An adhesion structure comprising titanium is formed over the buffer structure, wherein the adhesion structure has an adhesion sidewall thickness. An electromagnetic reflection structure comprising titanium nitride is formed over the adhesion structure, wherein the electromagnetic reflection structure has a reflection sidewall thickness greater than or equal to three times the adhesion sidewall thickness. A fill structure comprising a second oxide layer is formed that covers a top surface of the adhesion structure, a top surface of the

electromagnetic reflection structure, and an inner surface of the electromagnetic reflection structure.

FIGS. 1A-1T are illustrations of a semiconductor arrangement **100** at various stages of fabrication, according to some embodiments. FIGS. 1A-1T illustrate cross-sectional views of the semiconductor arrangement **100**. The semiconductor arrangement **100** comprises a semiconductor substrate **102**, an interconnect structure **104**, and a second substrate **106**. In some embodiments, the semiconductor substrate **102** corresponds to a device wafer of the semiconductor arrangement **100** and the second substrate **106** to a carrier wafer of the semiconductor arrangement **100**. The semiconductor substrate **102** has a top side **108** and a bottom side **110**, where the top side **108** corresponds to a first side of the semiconductor substrate **102** and the bottom side **110** corresponds to a second side of the semiconductor substrate **102**.

The semiconductor substrate **102** comprises at least one of an epitaxial layer, a silicon-on-insulator (SOI) structure, a wafer, or a die formed from a wafer. For example, the epitaxial layer may comprise a deposition of an overlayer on a crystalline substrate, where the overlayer is in registry with the semiconductor substrate **102**. In some embodiments, the epitaxial layer may comprise a type of crystal growth or material deposition in which new crystalline layers are formed with one or more well-defined orientations with respect to a crystalline seed layer of the semiconductor substrate **102**. The semiconductor substrate **102** comprises at least one of silicon, germanium, carbide, arsenide, gallium, arsenic, phosphide, indium, antimonide, SiGe, SiC, GaAs, GaN, GaP, InGaP, InP, InAs, InSb, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP or other suitable material. The semiconductor substrate **102** comprises at least one of monocrystalline silicon, crystalline silicon with a <100> crystallographic orientation, crystalline silicon with a <110> crystallographic orientation, crystalline silicon with a <111> crystallographic orientation or other suitable material.

In some embodiments, the semiconductor substrate **102** has at least one of a doped region **102a**, such as an n-type doped region or a p-type doped region, and a second region **102b**, such as a non-doped region or a different type of doped region than the doped region **102a**. An n-type doped region may be formed by doping the semiconductor substrate **102** with an electron donor element. In n-type semiconductors, electrons are majority charge carriers and holes are minority charge carriers. In an n-type doped region, the Fermi level is greater than the Fermi level of the intrinsic semiconductor base material and lies closer to the conduction band than the valence band. In some embodiments, a dopant for n-type silicon comprises phosphorus, arsenic, or other material that creates electrons as majority charge carriers. A p-type doped region may be formed by doping the semiconductor substrate **102** with an electron acceptor element. In p-type semiconductors, positive charges, also known as electron holes, are the majority charge carriers and electrons are the minority charge carriers. In some embodiments, p-type semiconductors may have a larger hole concentration than electron concentration. In a p-type doped region, the Fermi level is below the intrinsic semiconductor base material and lies closer to the valence band than the conduction band. In some embodiments, a dopant for p-type silicon is boron, gallium, or other material that creates electron holes as majority charge carriers. Other structures and/or configurations of the semiconductor substrate **102** are within the scope of the present disclosure.

In some embodiments, the semiconductor arrangement **100** comprises semiconductor devices **112** in the semiconductor substrate **102**, such as first semiconductor device

112a, second semiconductor device **112b**, and third semiconductor device **112c**, as illustrated. The semiconductor devices **112** are formed by at least one of doping, ion implantation, molecular diffusion, or other suitable techniques. In some embodiments, the semiconductor devices **112** are configured to sense radiation, such as incident light, which is projected towards the semiconductor substrate **102** along direction **113**. In some embodiments, the semiconductor devices **112** comprise photodiodes, such as at least one of pinned layer photodiodes, phototransistors, photogates, or other suitable components. In some embodiments, the semiconductor devices **112** are formed before, during, and/or after formation of other structures or components, described in greater detail herein. At least some of semiconductor devices **112** can vary from one another to have at least one of different heights, thicknesses, widths, material compositions, etc. Any number of semiconductor devices **112** in the semiconductor substrate **102** are contemplated.

At least some of the semiconductor devices **112** comprise at least one of germanium, indium, phosphorous, BF₂, arsenic, antimony, fluorine, InAs, InSb, GaSb, GaAs, InP, a silicide, or other suitable material. In some embodiments, the semiconductor devices **112** are configured to sense radiation, such as incident light, which is projected towards the semiconductor substrate **102**. The sensed radiation may be projected along the direction **113**, at an angle to the direction **113**, or a combination thereof. The sensed radiation may be incident at the top side **108** of the semiconductor substrate **102** before being sensed by one of the semiconductor devices **112**. At least some of the semiconductor devices **112** can comprise a material that is relatively highly absorptive to optical wavelengths of radiation. Optical radiation, also known as visible radiation, may comprise wavelengths from about 420 nanometers (nm) to about 680 nm or from about 380 nm to about 800 nm. White radiation is generally a combination of all optical radiation. According to some embodiments, optical radiation comprises red radiation (e.g., generally a red range of wavelengths from about 620 nm to about 700 nm), green radiation (e.g., generally a green range of wavelengths from about 492 nm to about 577 nm), blue radiation (e.g., generally a blue range of wavelengths from about 455 nm to about 492 nm), and white radiation (e.g., generally a white range of wavelengths from about 380 nm to about 800 nm). At least some of the semiconductor devices **112** may comprise a material that is relatively highly absorptive to near-infrared (NIR) wavelengths, such as radiation having a wavelength range from about 750 nm to about 1400 nm or may comprise an NIR range of wavelengths from about 780 nm to about 2500 nm. Other structures and/or configurations of the semiconductor devices **112** are within the scope of the present disclosure.

The interconnect structure **104** comprises one or more interconnect layers, such as at least one of a first interconnect layer **114**, a second interconnect layer **116**, a third interconnect layer **118**, or a fourth interconnect layer **120**. The one or more interconnect layers of the interconnect structure **104** comprise patterned dielectric layers and/or conductive layers that provide interconnections, such as wiring, between at least one of various doped features, circuitry, input/output, etc. of the semiconductor arrangement **100**. In some embodiments, the interconnect structure **104** comprises an interlayer dielectric and multilayer interconnect structures, such as at least one of contacts, vias, metal lines, or other type of structure. Other structures and configurations of the interconnect structure **104** are within the scope of the present disclosure. For purposes of illustration, the interconnect structure **104** comprises conductive

lines, where the positioning and configuration of such conductive lines might vary depending upon design needs. The interconnect structure **104** at least one of overlies the semiconductor substrate **102**, is in direct contact with the semiconductor substrate **102**, or is in indirect contact with the semiconductor substrate **102**.

The second substrate **106** comprises at least one of an epitaxial layer, a SOI structure, a wafer, or a die formed from a wafer. The second substrate **106** comprises at least one of silicon, germanium, carbide, arsenide, gallium, arsenic, phosphide, indium, antimonide, SiGe, SiC, GaAs, GaN, GaP, InGaP, InP, InAs, InSb, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP or other suitable material. The second substrate **106** comprises at least one of monocrystalline silicon, crystalline silicon with a <100> crystallographic orientation, crystalline silicon with a <110> crystallographic orientation, crystalline silicon with a <111> crystallographic orientation or other suitable material. In some embodiments, the second substrate **106** comprises at least one doped region. Other structures and/or configurations of the second substrate **106** are within the scope of the present disclosure.

In some embodiments, the second substrate **106** is bonded with the interconnect structure **104**, such as by at least one of one or more bonding layers, an adhesive, a bonding process, or other suitable techniques. In some embodiments where the second substrate **106** is bonded with the interconnect structure **104** using the one or more bonding layers, the one or more bonding layers are between the second substrate **106** and the interconnect structure **104**. The second substrate **106** at least one of overlies the interconnect structure **104**, is in direct contact with the interconnect structure **104**, or is in indirect contact with the interconnect structure **104**.

In some embodiments, one or more components of the semiconductor arrangement **100** are formed through an inversion operation. An inversion operation may be performed such that the semiconductor substrate **102** lies beneath at least one of the interconnect structure **104** or the second substrate **106** during formation of one or more components, such as the semiconductor devices **112**. For example, during formation of the interconnect structure **104**, the top side **108** of the semiconductor substrate **102** corresponds to a back side of the semiconductor substrate **102**, and the bottom side **110** of the semiconductor substrate **102** corresponds to a front side of the semiconductor substrate **102**. In some embodiments, a portion of the semiconductor substrate **102** on the top side **108** of the semiconductor substrate **102** is removed, such as after the inversion operation, to reduce a thickness of the semiconductor substrate **102**.

FIG. 1B illustrates a mask layer **124** formed over the semiconductor substrate **102**, according to some embodiments. The mask layer **124** at least one of overlies the semiconductor substrate **102**, is in direct contact with the semiconductor substrate **102**, or is in indirect contact with the semiconductor substrate **102**. In some embodiments, the mask layer **124** is a hard mask layer. The mask layer **124** comprises at least one of oxide, nitride, a metal, or other suitable material. The mask layer **124** is formed by at least one of physical vapor deposition (PVD), sputtering, chemical vapor deposition (CVD), low pressure CVD (LPCVD), atomic layer chemical vapor deposition (ALCVD), ultrahigh vacuum CVD (UHV CVD), reduced pressure CVD (RPCVD), atomic layer deposition (ALD), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), spin on, growth,

or other suitable techniques. Other structures and/or configurations of the mask layer **124** are within the scope of the present disclosure.

FIG. 1C illustrates the mask layer **124** patterned to form a patterned mask layer **126** over the semiconductor substrate **102**, according to some embodiments. According to some embodiments, a photoresist (not shown) is used to form the patterned mask layer **126**. The photoresist is formed over the mask layer **124** by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHV CVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. The photoresist comprises a light-sensitive material, where properties, such as solubility, of the photoresist are affected by light. The photoresist is a negative photoresist or a positive photoresist. With respect to a negative photoresist, regions of the negative photoresist become insoluble when illuminated by a light source, such that application of a solvent to the negative photoresist during a subsequent development stage removes non-illuminated regions of the negative photoresist. A pattern formed in the negative photoresist is thus a negative image of a pattern defined by opaque regions of a template, such as a mask, between the light source and the negative photoresist. In a positive photoresist, illuminated regions of the positive photoresist become soluble and are removed via application of a solvent during development. Thus, a pattern formed in the positive photoresist is a positive image of opaque regions of the template, such as a mask, between the light source and the positive photoresist. One or more etchants have a selectivity such that the one or more etchants remove or etch away one or more layers exposed or not covered by the photoresist at a greater rate than the one or more etchants remove or etch away the photoresist. Accordingly, an opening in the photoresist allows the one or more etchants to form a corresponding opening in the one or more layers under the photoresist, and thereby transfer a pattern in the photoresist to the one or more layers under the photoresist. The photoresist is stripped or washed away after the pattern transfer.

An etching process, used to remove portions of the mask layer **124** to form the patterned mask layer **126**, is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material. In some embodiments, the etching process, which is performed to remove portions of the mask layer **124** and form the patterned mask layer **126**, also removes at least some of the semiconductor substrate **102**, such as portions of the semiconductor substrate **102** underlying openings **128** in the patterned mask layer **126**. Other processes and/or techniques for forming the patterned mask layer **126** are within the scope of the present disclosure.

FIG. 1D illustrates use of the patterned mask layer **126** to form one or more recesses **130** in the semiconductor substrate **102**, according to some embodiments. In some embodiments, an etching process is performed to form the recesses **130**, where the openings **128** in the patterned mask layer **126** allow one or more etchants applied during the etching process to remove portions of the semiconductor substrate **102** while the patterned mask layer **126** protects or shields portions of the semiconductor substrate **102** that are covered by the patterned mask layer **126**. The etching process is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or

other suitable material. Other processes and/or techniques for forming the recesses **130** are within the scope of the present disclosure.

In some embodiments, one or more of the recesses **130** overlie a semiconductor device, such as one of the semiconductor devices **112**. Any number of recesses **130** over one of the semiconductor devices **112** are contemplated. In some embodiments, a portion of the semiconductor substrate **102** remains over one of the semiconductor devices **112** to separate the recess **130** therefrom. Other structures and/or configurations of the recesses **130** are within the scope of the present disclosure.

FIG. 1E illustrates removal of the patterned mask layer **126**, according to some embodiments. The patterned mask layer **126** is removed after the recesses **130** are formed. The patterned mask layer **126** is removed by at least one of chemical-mechanical polishing (CMP), etching, or other suitable techniques. The etching process is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material. Other processes and/or techniques for removing the patterned mask layer **126** are within the scope of the present disclosure.

In some embodiments, a portion of the semiconductor substrate **102** defines a recess **132** of the recesses **130**. The recess **132** has at least one of a first recess tapered sidewall **134** or a second recess tapered sidewall **136**. The first recess tapered sidewall **134** has a first recess slope, such as a negative slope, or the second recess tapered sidewall **136** has a second recess slope, such as a positive slope. In some embodiments, the second recess slope is opposite in polarity relative to the first recess slope. In some embodiments, the recess **132** has a triangular shape. In some embodiments, a cross-sectional area of the recess **132** decreases along the direction **113**, such that a width of an upper portion of the recess **132** is greater than a width of a lower portion of the recess **132**. Other structures and/or configurations of the recesses **130**, such as the recess **132**, are within the scope of the present disclosure.

In some embodiments, the semiconductor substrate **102** having a specific crystallographic orientation, such as crystalline silicon with at least one of a <100> crystallographic orientation, a <110> crystallographic orientation, or a <111> crystallographic orientation, enables an etching process to form the recess tapered sidewalls (e.g., the first recess tapered sidewall and the second recess tapered sidewall **136**). In some embodiments, portions of the semiconductor substrate **102** have different crystallographic orientations, such as at least one of a <100> crystallographic orientation, a <110> crystallographic orientation, or a <111> crystallographic orientation, where etch rates of the etching process differ between the different crystallographic orientations at least due to different densities of the different crystallographic orientations, resulting in the recessed tapered sidewalls being formed by the etching process.

In some embodiments, a first portion of the semiconductor substrate **102** having the first recess tapered sidewall **134** and the second recess tapered sidewall **136** has a first crystallographic orientation such as <111> crystallographic orientation, and a second portion of the semiconductor substrate **102** that is removed to form the recess **132** has a second crystallographic orientation such as <100> crystallographic orientation. In some embodiments, a first density of the semiconductor substrate **102**, such as a surface density of the first crystallographic orientation, is greater than a

second density of the semiconductor substrate, such as a surface density of the second crystallographic orientation. Such differences in density may facilitate the etching process to remove the second portion of the semiconductor substrate **102** while removing little to none of the first portion of the semiconductor substrate **102** due to an etch rate of the second portion of the semiconductor substrate **102** being higher than an etch rate of the first portion of the semiconductor substrate **102**. Other processes and/or techniques for forming the recessed tapered sidewalls defining the recesses **130**, such as the recess **132**, are within the scope of the present disclosure.

In some embodiments, a device distance **138** between a top surface of a semiconductor device, such as the second semiconductor device **112b**, and at least one of an uppermost portion of the recess **132** or the top side **108** of the semiconductor substrate **102** is less than or equal to about 40,000 Å. In some embodiments, a recess distance **140** between two adjacent recesses of the recesses **130** is between about zero Å to about 20,000 Å.

In some embodiments, a set of recesses **141** are adjacent each other, such as in an offset saw tooth configuration. In some embodiments, at least some recesses of one or more sets of recesses overlie one of the semiconductor devices **112**. In some embodiments, the set of recesses **141** are at least one of vertically coincident with one of the semiconductor devices **112**, vertically coincident with a portion of one of the semiconductor devices **112**, or vertically offset from one of the semiconductor devices **112**. For example, as illustrated in FIG. 1E, the set of recesses **141** are vertically coincident with the third semiconductor device **112c**. In some embodiments, the set of recesses **141** are at least one of surrounded by a doped region of the semiconductor substrate **102**, surrounded by a portion of a doped region of the semiconductor substrate **102**, laterally surrounded by a doped region of the semiconductor substrate **102**, or laterally surrounded by a portion of a doped region of the semiconductor substrate **102**. For example, as illustrated in FIG. 1E, the set of recesses **141** are laterally surrounded by the doped region **102a** of the semiconductor substrate **102** and are surrounded by a portion of the doped region **102a** of the semiconductor substrate **102** that underlies the set of recesses **141**. Other structures and/or configurations of the recesses **141** are within the scope of the present disclosure.

FIG. 1F illustrates a cross-sectional view of the semiconductor arrangement **100**, according to some embodiments, where at least some recesses **130** are directly adjacent. In some embodiments, a second set of recesses **142** are directly adjacent each other, such as in a saw tooth configuration. In some embodiments, at least some recesses of one or more second sets of recesses overlie one of the semiconductor devices **112**. In some embodiments, the second set of recesses **142** are at least one of vertically coincident with one of the semiconductor devices **112**, vertically coincident with a portion of one of the semiconductor devices **112**, or vertically offset from one of the semiconductor devices **112**. For example, as illustrated in FIG. 1F, the second set of recesses **142** are vertically coincident with the second semiconductor device **112b**. In some embodiments, the second set of recesses **142** are at least one of surrounded by a doped region of the semiconductor substrate **102**, surrounded by a portion of a doped region of the semiconductor substrate **102**, laterally surrounded by a doped region of the semiconductor substrate **102**, or laterally surrounded by a portion of a doped region of the semiconductor substrate **102**. For example, as illustrated in FIG. 1F, the second set of recesses **142** are laterally surrounded by the doped region **102a** of the

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semiconductor substrate **102** and are surrounded by a portion of the doped region **102a** of the semiconductor substrate **102** that underlies the second set of recesses **142**.

FIG. 1G illustrates a first dielectric layer **144** formed over the semiconductor substrate **102**, according to some embodiments. In some embodiments, the first dielectric layer **144** is in direct contact with the top side **108** of the semiconductor substrate **102** and/or the recess tapered sidewalls defined in the semiconductor substrate **102**, such as the first recess tapered sidewall **134** and the second recess tapered sidewall **136** defining the recess **132**. In some embodiments, the first dielectric layer **144** is in indirect contact with the top side **108** of the semiconductor substrate **102** and/or the recess tapered sidewalls defined in the semiconductor substrate **102**. Other structures and/or configurations of the first dielectric layer **144** are within the scope of the present disclosure.

In some embodiments, the semiconductor arrangement **100** comprises a buffer layer (not shown) between the semiconductor substrate **102** and the first dielectric layer **144**, such as formed the semiconductor substrate **102** prior to forming the first dielectric layer **144**. The buffer layer is in direct contact with the top side **108** of the semiconductor substrate **102** and/or the recess tapered sidewalls defined in the semiconductor substrate **102**, or is in indirect contact with the top side **108** of the semiconductor substrate **102** and/or the recess tapered sidewalls defined in the semiconductor substrate **102**.

The buffer layer comprises at least one of an anti-reflection coating, SiO_2 , HfSiON , HfSiO_x , HfAlO_x , HfO_2 , ZrO_2 , La_2O_3 , Y_2O_3 , or other suitable material. The buffer layer is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, the buffer layer comprises a single layer that is configured to provide adhesion between the first dielectric layer **144** and the semiconductor substrate **102**. According to some embodiments, the buffer layer comprises multiple layers, where an outer layer of the multiple layers is configured to provide adhesion with the first dielectric layer **144**. When the semiconductor arrangement **100** comprises the buffer layer, the first dielectric layer **144** at least one of overlies the buffer layer, is in direct contact with a top surface of the buffer layer, or is in indirect contact with the top surface of the buffer layer. Other structures and/or configurations of the buffer layer are within the scope of the present disclosure.

The first dielectric layer **144** comprises at least one of SiO , SiO_2 , SiN , Si_3N_4 , MgO , Al_2O_3 , Yb_2O_3 , ZnO , Ta_2O_5 , ZrO_2 , HfO_2 , TeO_2 , TiO_2 , an oxide layer, or other suitable material. The first dielectric layer **144** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. The first dielectric layer **144** is formed over at least one of in the recesses **130**, a set of the recesses **130**, or all of the recesses **130**. In some embodiments, the first dielectric layer **144** is formed over the top side **108** of the semiconductor substrate **102**. In some embodiments, a first dielectric layer distance **146** between a top surface of the first dielectric layer **144** and the top side **108** of the semiconductor substrate **102** is less than or equal to about 10,000 Å.

In some embodiments, dielectric layer portions of the first dielectric layer **144** are in at least one of the recesses **130**. For example, as illustrated in FIG. 1G, a first dielectric layer portion **148** of the first dielectric layer **144** is in the recess **132**. The first dielectric layer portion **148** of the first dielectric layer **144** has a first dielectric tapered sidewall **150** with

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which the first recess tapered sidewall **134** of the semiconductor substrate **102** aligns. When the semiconductor arrangement **100** comprises the buffer layer over the semiconductor substrate **102**, a portion of the buffer layer separates the first dielectric tapered sidewall **150** of the first dielectric layer portion **148** of the first dielectric layer **144** from the first recess tapered sidewall **134** of the semiconductor substrate **102**.

The first dielectric layer portion **148** of the first dielectric layer **144** has a second dielectric tapered sidewall **152** with which the second recess tapered sidewall **136** of the semiconductor substrate **102** aligns. When the semiconductor arrangement **100** comprises the buffer layer over the semiconductor substrate **102**, a portion of the buffer layer separates the second dielectric tapered sidewall **152** of the first dielectric layer portion **148** of the first dielectric layer **144** from the second recess tapered sidewall **136** of the semiconductor substrate **102**. In some embodiments, one or more of the dielectric layer portions of the first dielectric layer **144** overlies one of the semiconductor devices **112**. For example, as illustrated in FIG. 1G, the first dielectric layer portion **148** overlies the first semiconductor device **112a**. In some embodiments, one or more of the dielectric layer portions of the first dielectric layer **144** are at least one of vertically coincident with one of the semiconductor devices **112**, vertically coincident with a portion of one of the semiconductor devices **112**, or vertically offset from one of the semiconductor devices **112**. For example, as illustrated in FIG. 1G, the first dielectric layer portion **148** is vertically coincident with the first semiconductor device **112a**. In some embodiments, at least one of a portion of the buffer layer or a portion of the semiconductor substrate **102** separate the first dielectric layer portion **148** of the first dielectric layer **144** from the first semiconductor device **112a**.

In some embodiments, the first dielectric layer portion **148** of the first dielectric layer **144** in the recess **132** is a high absorption (HA) structure, such as due, at least in part, to at least one of the first dielectric tapered sidewall **150**, the second dielectric tapered sidewall **152**, the first recess tapered sidewall **134**, or the second recess tapered sidewall **136**. An HA structure, such as the first dielectric layer portion **148**, directs more radiation to a semiconductor device underlying the first dielectric layer portion **148** of the first dielectric layer **144** as compared to a portion of the first dielectric layer **144** and a portion of the semiconductor substrate **102** that do not have one or more tapered sidewalls. One or more additional portions of the first dielectric layer **144** in recesses **130** in the semiconductor substrate **102** are similarly constructed HA structures that overlie one of the semiconductor devices **112**.

In some embodiments, two or more HA structures are laterally adjacent each other. For example, as illustrated in FIG. 1G, a first HA structure **154** is laterally adjacent to a second HA structure **156**. An HA distance **158** between two adjacent HA structures, such as the first HA structure **154** and the second HA structure **156** is between about zero Å to about 20,000 Å. In some embodiments, two or more HA structures **710** are directly adjacent each other. For example, as illustrated in FIG. 1G, a set of HA structures **160** are directly adjacent each other, such as in a saw tooth configuration. In some embodiments, at least some HA structures of one or more sets of HA structures are at least one of vertically coincident with one of the semiconductor devices **112**, vertically coincident with a portion of one of the semiconductor devices **112**, or vertically offset from one of the semiconductor devices **112**. For example, as illustrated in FIG. 1G, the set of HA structures **160** are vertically

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coincident with the third semiconductor device **112c**. In some embodiments, at least some HA structures of one or more sets of HA structures are at least one of surrounded by a doped region of the semiconductor substrate **102**, surrounded by a portion of a doped region of the semiconductor substrate **102**, laterally surrounded by a doped region of the semiconductor substrate **102**, or laterally surrounded by a portion of a doped region of the semiconductor substrate **102**. For example, as illustrated in FIG. 1G, the set of HA structures **160** are laterally surrounded by the doped region **102a** of the semiconductor substrate **102** and are surrounded by a portion of the doped region **102a** of the semiconductor substrate **102** that underlies the set of HA structures **160**. Other structures and/or configurations of HA structures are within the scope of the present disclosure.

FIG. 1H illustrates a photoresist **162** formed over the first dielectric layer **144**, according to some embodiments. The photoresist **162** at least one of overlies the first dielectric layer **144**, is in direct contact with the first dielectric layer **144**, or is in indirect contact with the first dielectric layer **144**. The photoresist **162** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. The photoresist **162** comprises a light-sensitive material, where properties, such as solubility, of the photoresist **162** are affected by light. The photoresist **162** is a negative photoresist or a positive photoresist.

FIG. 1I illustrates the photoresist **162** patterned to form a patterned photoresist **164** over the first dielectric layer **144**, according to some embodiments. The patterned photoresist **164** has openings exposing portions of the first dielectric layer **144**. In some embodiments, the openings in the patterned photoresist **164** are between the semiconductor devices **112**, such that the openings do not overlie or are laterally offset from the semiconductor devices **112**. In some embodiments, an opening in the patterned photoresist **164** is between two adjacent semiconductor devices, such that the opening overlies a portion of the semiconductor substrate **102** between, for example, the first semiconductor device **112a** and the second semiconductor device **112b**. According to some embodiments, an opening in the patterned photoresist **164** overlies a portion of one or more of the semiconductor devices **112**.

FIG. 1J illustrates shallow trenches **166** formed using the patterned photoresist **164**, according to some embodiments. The shallow trenches **166** extend through the first dielectric layer **144** and into the semiconductor substrate **102**. In some embodiments, the shallow trenches **166** extend into the doped region **102a** of the semiconductor substrate **102**. In some embodiments, the shallow trenches **166** extend into the second region **102b** of the semiconductor substrate **102**. The shallow trenches **166** are at least one of laterally offset from a semiconductor device or between two of the semiconductor devices **112**. In some embodiments, one of the shallow trenches **166**, such as a shallow trench **168**, is between two adjacent semiconductor devices, such as the first semiconductor device **112a** and the second semiconductor device **112b**. In some embodiments, one of the shallow trenches **166**, such as the second shallow trench **170**, is adjacent to the shallow trench **168**. For example, as illustrated in FIG. 1J, the second shallow trench **170** is adjacent to the shallow trench **168** with the second semiconductor device **112b** disposed therebetween. In some embodiments, one or more portions of the semiconductor substrate **102** separates one of the shallow trenches **166** from one or more of the semiconductor devices **112**. For example, as illustrated in FIG. 1J, a first semiconductor substrate portion **172** of the semicon-

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ductor substrate **102** separates the shallow trench **168** from the first semiconductor device **112a** of two adjacent semiconductor devices (the first semiconductor device **112a** and the second semiconductor device **112b**). A second semiconductor substrate portion **174** of the semiconductor substrate **102** separates the shallow trench **168** from the second semiconductor device **112b** of two adjacent semiconductor devices (the first semiconductor device **112a** and the second semiconductor device **112b**). In some embodiments, an etching process is performed to form the shallow trenches **166**, where openings in the patterned photoresist **164** allow one or more etchants applied during the etching process to remove portions of the first dielectric layer **144** and/or portions of the semiconductor substrate **102** while the patterned photoresist **164** protects or shields portions of the first dielectric layer **144** and/or portions of the semiconductor substrate **102** that are covered by the patterned photoresist **164**. The etching process is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material.

In some embodiments, the shallow trench **168** has at least one of a first trench tapered sidewall **176** or a second trench tapered sidewall **178**. The first trench tapered sidewall **176** has a first trench slope, such as a negative slope, or the second trench tapered sidewall **178** has a second trench slope, such as a positive slope. In some embodiments, the second trench slope is opposite in polarity relative to the first trench slope. In some embodiments, the shallow trench **168** has a trapezoidal shape such that a third trench sidewall **180** connects the first trench tapered sidewall **176** to the second trench tapered sidewall **178**. In some embodiments, the third trench tapered sidewall is perpendicular to the direction **113**. In some embodiments, a cross-sectional area of the shallow trench **168** decreases along the direction **113**, such that a width of an upper portion of the shallow trench **168** is greater than a width of a lower portion of the shallow trench **168**.

According to some embodiments, the first semiconductor device **112a** or portions of the first semiconductor device **112a** are adjacent to or directly adjacent to the first trench tapered sidewall **176**. In some embodiments, the second semiconductor device **112b** or portions of the second semiconductor device **112b** are adjacent to or directly adjacent to the second trench tapered sidewall **178**. The doped region **102a** of the semiconductor substrate **102** has a top doped side **182** and a bottom doped side **184**. In some embodiments, the top doped side **182** of the doped region **102a** of the semiconductor substrate **102** is contiguous with the top side **108** of the semiconductor substrate **102**. In some embodiments, the top doped side **182** of the doped region **102a** of the semiconductor substrate **102** is vertically offset from the top side **108** of the semiconductor substrate **102**. In some embodiments, a bottom side **186** of the first semiconductor device **112a** or a portion of the bottom side **186** of the first semiconductor device **112a** is adjacent or directly adjacent to the bottom doped side **184** of the doped region **102a** of the semiconductor substrate **102**.

In some embodiments, the third trench sidewall **180** extends a third trench sidewall distance **188** into the semiconductor substrate **102** from the top side **108** of the semiconductor substrate **102**. In some embodiments, the bottom side **186** of the first semiconductor device **112a** extends a bottom side distance **190** into the semiconductor substrate **102** from the top side **108** of the semiconductor substrate **102**. In some embodiments, the third trench side-

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wall distance **188** is greater than the bottom side distance **190**. In some embodiments, the doped region **102a** of the semiconductor substrate **102** extends a doped region distance **192** into the semiconductor substrate **102** from the top side **108** of the semiconductor substrate **102**. In some

embodiments, the third trench sidewall distance **188** is greater than the doped region distance **192**. Other processes and/or techniques for forming the shallow trenches **166** are within the scope of the present disclosure.

FIG. 1K illustrates removal of the patterned photoresist **164**, according to some embodiments. The patterned photoresist **164** is removed after the shallow trenches **166** are formed. The patterned photoresist **164** is removed by at least one of CMP, etching, or other suitable techniques. In some

embodiments, removal of the patterned photoresist **164** exposes the top surface of the first dielectric layer **144**.

A portion of the first dielectric layer **144** defining the shallow trench **168** has a first dielectric sidewall **194** and a second dielectric sidewall **196**. In some embodiments, at least some of the first dielectric sidewall **194** is tapered and/or at least some of the second dielectric sidewall **196** is tapered. The first dielectric sidewall **194** has a first slope, such as a negative slope, and/or the second dielectric sidewall **196** has a second slope, such as a positive slope. In some embodiments, the second slope is opposite in polarity relative to the first slope. In some embodiments, a cross-sectional area of the first dielectric layer **144** defining the shallow trench **168** decreases along the direction **113**, such that a width of an upper portion of the shallow trench **168** is greater than a width of a lower portion of the shallow trench **168**.

According to some embodiments, at least some of a sidewall defining the shallow trench **168**, such as at least some of the first trench tapered sidewall **176**, at least some of the second trench tapered sidewall **178**, at least some of the first dielectric sidewall **194**, and/or at least some of the second dielectric sidewall **196**, extends vertically, such as in a direction parallel to the direction **113**.

In some embodiments, a lowermost portion of the shallow trench **168** (e.g., the third trench sidewall **180** of the shallow trench **168**) is lower than an uppermost portion of one of the semiconductor devices **112** (e.g., a top side **198** of the first semiconductor device **112a**). According to some embodiments, the lowermost portion of the shallow trench **168** is lower than a lowermost portion of one of the semiconductor devices **112** (e.g., the bottom side **186** of the first semiconductor device **112a**). According to some embodiments, the lowermost portion of the shallow trench **168** is lower than lowermost portions of adjacent semiconductor devices (e.g., the bottom side **186** of the first semiconductor device **112a** and a bottom side **200** of the second semiconductor device **112b**). Other structures and/or configurations of the shallow trenches **166** are within the scope of the present disclosure.

FIG. 1L illustrates a buffer layer **202** formed over the first dielectric layer **144** and in the shallow trenches **166**, according to some embodiments. In some embodiments, the buffer layer **202** is in direct contact with the top surface of the first dielectric layer **144**, the first trench tapered sidewall **176**, the second trench tapered sidewall **178**, and/or the third trench sidewall **180** defined in the semiconductor substrate **102**. In some embodiments, the buffer layer **202** is in direct contact with the first dielectric sidewall **194** and/or the second dielectric sidewall **196** defined in the first dielectric layer **144**. In some embodiments, the buffer layer **202** is in indirect contact with the top surface of the first dielectric layer **144**, the first trench tapered sidewall **176**, the second trench tapered sidewall **178**, and/or the third trench sidewall **180**

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defined in the semiconductor substrate **102**. In some embodiments, the buffer layer **202** is in indirect contact with the first dielectric sidewall **194** and/or the second dielectric sidewall **196** defined in the first dielectric layer **144**.

In some embodiments, the buffer layer **202** comprises at least one of SiO, SiO₂, HfSiON, HfSiO_x, HfAlO_x, HfO₂, ZrO₂, La₂O₃, Y₂O₃, silicon nitride (Si₃N₄), silicon oxynitride (SiO_xN_y), aluminum oxide, hafnium oxide, tantalum nitride, spin-on glass (SOG), fluorinated silica glass (FSG), carbon doped silicon oxide (e.g., SiCOH), a high-density-plasma (HDP) oxide, an HDP-CVD oxide, or other suitable material. The buffer layer **202** is formed by at least one of PVD, sputtering, CVD, HDP-CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. For example, in HDP-CVD, a gas mixture comprising silicon, oxygen, and one or more inert gases may be used. The one or more inert gases of this mixture may comprise about less than 20% of the total gas mixture. In some embodiments, the gas mixture may comprise silane, oxygen, and an inert gas, such as argon. The inert gas pressure and chamber pressure are kept low while the sputter rf energy is increased.

A first portion of the buffer layer **202** is in the shallow trench **168**. In some embodiments, the first portion of the buffer layer **202** has a first buffer sidewall **204** with which at least one of the first trench tapered sidewall **176** of the semiconductor substrate **102** or the first dielectric sidewall **194** of the first dielectric layer **144** aligns. The first portion of the buffer layer **202** in the shallow trench **168** has a second buffer sidewall **206** with which at least one of the second trench tapered sidewall **178** of the semiconductor substrate **102** or the second dielectric sidewall **196** of the first dielectric layer **144** aligns. In some embodiments, the first portion of the buffer layer **202** in the shallow trench **168** has a third buffer sidewall **208** with which the third trench sidewall **180** of the semiconductor substrate **102** aligns.

In some embodiments, a portion of the buffer layer **202** comprises a buffer structure, such as one or more portions of a buffer sidewall (e.g., the first buffer sidewall **204**, the second buffer sidewall **206**, the third buffer sidewall **208**, etc.), which has a buffer sidewall thickness **210**. In some embodiments, the buffer sidewall thickness **210** is between about (≈) 100 Å to about 500 Å to buffer against contamination, such as dopants of the doped region **102a**. Such contamination may also be present from manufacturing operations, such as a cleaning process, which may present chemicals that could damage, react with, and/or penetrate inside components or devices in the semiconductor substrate **102**. The sidewall thickness **210** is not merely a matter of design choice, but is critical to, at least, provide sufficient buffering of contamination while not consuming too much space of a shallow trench within which the buffer layer **202** is formed. If the sidewall thickness **210** is too thin, for example, then contaminants might pass through the buffer layer **202** and/or the buffer layer **202** might not provide sufficient stress release. If the buffer layer **202** is too thick, for example, then there might not be sufficient room in the shallow trench within which the buffer layer **202** is formed to accommodate additional layers, materials, etc. necessary for desired device operation, functioning, etc. If the buffer layer **202** is too thick, for example, then subsequent layers formed on the first buffer sidewall **204**, the second buffer sidewall **206**, etc., such as an adhesion layer and a reflection layer (set forth in greater detail below) might be too close to provide proper adhesion and/or reflection. If the buffer layer **202** is too thick, for example, then efficacy of etching, cleaning, etc. of the shallow trench during subsequent opera-

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tions may be reduced causing improper formation of subsequent layers, contamination, etc. Other structures and/or configurations of the buffer layer **202** are within the scope of the present disclosure.

FIG. 1M illustrates removal of portions of the buffer layer **202** and portions of the first dielectric layer **144**, according to some embodiments. In some embodiments, portions of the buffer layer **202** along top surfaces of first dielectric layer **144** and along sidewalls of the first dielectric layer **144** (e.g., the first dielectric sidewall **194** and the second dielectric sidewall **196**) are removed to retain buffer structures, such as a buffer structure **212**. In some embodiments, the buffer structure **212** is contiguous with the doped region **102a** of the semiconductor substrate **102**. The buffer structure **212** has the buffer sidewall thickness **210**, set forth in greater detail herein. In some embodiments, portions of the first dielectric layer **144** above the top side **108** of the semiconductor substrate **102** are removed to retain dielectric layer portions within the recesses **130**, such as the first dielectric layer portion **148** within the recess **132**. The portions of the buffer layer **202** and the portions of the first dielectric layer **144** are removed by at least one of chemical-mechanical polishing (CMP), etching, or other suitable techniques. The etching process is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material.

In some embodiments, the buffer structures, such as the buffer structure **212**, are at least one of horizontally coincident with one or more of the semiconductor devices **112**, horizontally coincident with a portion of one or more of the semiconductor devices **112**, horizontally offset from one or more of the semiconductor devices **112**, or horizontally offset from a portion of one or more of the semiconductor devices **112**. For example, as illustrated in FIG. 1M, the buffer structure **212** is horizontally offset from a portion of the first semiconductor device **112a** and horizontally offset from a portion of the second semiconductor device **112b**. As set forth herein and in some embodiments, one or more of the semiconductor devices **112** or portions of the semiconductor devices **112** may be coincident with the tapered sidewalls of the shallow trenches **166** such that portions of the semiconductor devices **112** are horizontally offset from the buffer structure **212** that aligns with the tapered sidewalls.

In some embodiments, the buffer structure **212** comprises at least one of the first buffer sidewall **204** or the second buffer sidewall **206**. The first buffer sidewall **204** has a first buffer slope, such as a negative slope, or the second buffer sidewall **206** has a second buffer slope, such as a positive slope. In some embodiments, the second buffer slope is opposite in polarity relative to the first buffer slope. In some embodiments, the buffer structure **212** has a trapezoidal shape such that the third buffer sidewall **208** connects the first buffer sidewall **204** to the second buffer sidewall **206**. In some embodiments, the third buffer sidewall **208** is perpendicular to the direction **113**. In some embodiments, a cross-sectional area of the buffer structure **212** decreases along the direction **113**, such that a width of an upper portion of the buffer structure **212** is greater than a width of a lower portion of the buffer structure **212**. Other processes and/or techniques for removing the portions of the buffer layer **202** and the portions of the first dielectric layer **144** are within the scope of the present disclosure.

FIG. 1N illustrates an adhesion layer **214** formed over the semiconductor substrate **102** and in the shallow trenches

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166, according to some embodiments. In some embodiments, the adhesion layer **214** is in direct contact with the top side **108** of the semiconductor substrate **102**, the first buffer sidewall **204**, the second buffer sidewall **206**, and/or the third buffer sidewall **208** of the buffer structure **212**. In some embodiments, the adhesion layer **214** is in indirect contact with the top side **108** of the semiconductor substrate **102**, the first buffer sidewall **204**, the second buffer sidewall **206**, and/or the third buffer sidewall **208** of the buffer structure **212**. In some embodiments, the adhesion layer **214** is formed as a single layer in a single manufacturing operation. In some embodiments, the adhesion layer **214** is formed as multiple layers of the same material in multiple manufacturing operations.

In some embodiments, the adhesion layer **214** comprises titanium (Ti), a mixture of Ti and one or more metals, or other suitable material. The adhesion layer **214** is formed by at least one of PVD, sputtering, CVD, HDP-CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. A first portion of the adhesion layer **214** is in the shallow trench **168**. In some embodiments, the first portion of the adhesion layer **214** has a first adhesion sidewall **216** with which the first buffer sidewall **204** of the buffer structure **212** aligns. The first portion of the adhesion layer **214** in the shallow trench **168** has a second adhesion sidewall **218** with which the second buffer sidewall **206** of the buffer structure **212** aligns. In some embodiments, the first portion of the adhesion layer **214** in the shallow trench **168** has a third adhesion sidewall **220** with which the third buffer sidewall **208** of the buffer structure **212** aligns.

In some embodiments, a portion of the adhesion layer **214** comprises one or more adhesion structures. An adhesion structure, such as one or more portions of an adhesion sidewall (e.g., the first adhesion sidewall **216**, the second adhesion sidewall **218**, the third adhesion sidewall **220**, etc.), is, for example, adhesively bonded to the buffer structure **212** has an adhesion sidewall thickness **222**. In some embodiments, the adhesion sidewall thickness **222** is greater than about ($\geq$) 100 Å to provide adhesion to the buffer structure **212**. The adhesion sidewall thickness **222** is not merely a matter of design choice, but is critical to, at least, provide sufficient adhesion to the buffer structure **212**. If the adhesion sidewall thickness **222** is too thick and/or too thin, for example, then the adhesion layer **214** might not sufficiently bond to the buffer structure **212**. If the adhesion layer **214** is too thick, for example, then subsequent layers formed on the first adhesion sidewall **216**, the second adhesion sidewall **218**, etc., such as a reflection layer (set forth in greater detail below) might be too close to provide proper reflection. If the adhesion layer **214** is too thick, for example, then efficacy of etching, cleaning, etc. of the shallow trench during subsequent operations may be reduced causing improper formation of subsequent layers, contamination, etc. If the adhesion layer **214** is too thin, for example, then subsequent layers formed on the first adhesion sidewall **216**, the second adhesion sidewall **218**, etc., such as a reflection layer (set forth in greater detail below) might not sufficiently bond within the shallow trench. Other structures and/or configurations of the adhesion layer **214** are within the scope of the present disclosure.

FIG. 1O illustrates a reflection layer **224** formed over the semiconductor substrate **102**, over the adhesion layer **214**, and in the shallow trenches **166**, according to some embodiments. In some embodiments, the reflection layer **224** is in direct contact with a top side of the adhesion layer **214**, the first adhesion sidewall **216**, the second adhesion sidewall

218, and/or the third adhesion sidewall 220 of the adhesion layer 214. In some embodiments, the reflection layer 224 is in indirect contact with at least one of the top side of the adhesion layer 214, the top side 108 of the semiconductor substrate 102, the first adhesion sidewall 216, the first buffer sidewall 204, the first trench tapered sidewall 176, the second adhesion sidewall 218, the second buffer sidewall 206, the second trench tapered sidewall 178, the third adhesion sidewall 220, the third buffer sidewall 208, or the third trench sidewall 180. In some embodiments, the reflection layer 224 is formed as a single layer in a single manufacturing operation. In some embodiments, the reflection layer 224 is formed as multiple layers of the same material in multiple manufacturing operations.

In some embodiments, the reflection layer 224 comprises titanium nitride (TiN), a mixture of TiN and one or more metals, or other suitable material. The reflection layer 224 is formed by at least one of PVD, sputtering, CVD, HDP-CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. A first portion of the reflection layer 224 is in the shallow trench 168. In some embodiments, the first portion of the reflection layer 224 has a first reflection sidewall 226 with which the first adhesion sidewall 216 of the adhesion layer 214 aligns. The first portion of the reflection layer 224 in the shallow trench 168 has a second reflection sidewall 228 with which the second adhesion sidewall 218 of the adhesion layer 214 aligns. In some embodiments, the first portion of the reflection layer 224 in the shallow trench 168 has a third reflection sidewall 230 with which the third adhesion sidewall 220 of the adhesion layer 214 aligns.

In some embodiments, a portion of the reflection layer 224 comprises one or more reflection structures. A reflection structure, such as one or more portions of a reflection sidewall (e.g., the first reflection sidewall 226, the second reflection sidewall 228, the third reflection sidewall 230, etc.), for example, is adhesively bonded to one or more portions of the adhesion layer 214 (e.g., one or more portions of an adhesion structure) and has a reflection sidewall thickness 232. In some embodiments, the reflection sidewall thickness 232 is greater than or equal to three times the adhesion sidewall thickness 222. In some embodiments, the reflection sidewall thickness 232 is greater than or equal to about ($\geq$) 300 Å to provide reflection of electromagnetic radiation away from the shallow trenches 166 and towards the semiconductor devices 112 (e.g., a photodiode). The reflection sidewall thickness 232 (e.g., relative to the adhesion sidewall thickness 222) is not merely a matter of design choice, but is critical to, at least, provide sufficient reflection of electromagnetic radiation. If the reflection sidewall thickness 232 (e.g., relative to the adhesion sidewall thickness 222) is too thin, for example, then electromagnetic radiation might not be sufficiently reflected by the reflection structure, such that an insufficient amount of electromagnetic radiation might be reflected away from a shallow trench and towards a semiconductor device (e.g., and thus the semiconductor device might not register or provide an indication of electromagnetic radiation). If the reflection sidewall thickness 232 (e.g., relative to the adhesion sidewall thickness 222) is too thick, for example, then electromagnetic radiation might not be sufficiently reflected by the reflection structure, such that an insufficient amount of electromagnetic radiation might be reflected away from the shallow trench. If the reflection layer 224 is too thick, for example, then efficacy of etching, cleaning, etc. of the shallow trench during subsequent operations may be reduced causing improper formation of subsequent layers, contamination, etc. Other

structures and/or configurations of the reflection layer 224 are within the scope of the present disclosure.

FIG. 1P illustrates removal of portions of the adhesion layer 214 and portions of the reflection layer 224, according to some embodiments. In some embodiments, portions of the reflection layer 224 along top surfaces of adhesion layer 214 and along portions of the sidewalls of the adhesion layer 214 (e.g., a portion of the first reflection sidewall 226 and the second reflection sidewall 228) are removed to retain electromagnetic reflection structures, such as an electromagnetic reflection structure 234. In some embodiments, portions of the adhesion layer 214 along the top side 108 of the semiconductor substrate 102 and along portions of the sidewalls of the buffer structure 212 (e.g., a portion of the first adhesion sidewall 216 and the second adhesion sidewall 218) are removed to retain adhesion structures, such as an adhesion structure 236. In some embodiments, the adhesion structure 236 comprises titanium and is formed as an adhesion layer contiguous with the buffer structure 212. In some embodiments, the electromagnetic reflection structure 234 comprises titanium nitride and is formed as a reflection layer contiguous with the adhesion structure 236. In some embodiments, the electromagnetic reflection structure 234 has the reflection sidewall thickness 232 and the adhesion structure 236 has the adhesion sidewall thickness 222, set forth in greater detail herein. In some embodiments, the reflection sidewall thickness 232 is greater than the adhesion sidewall thickness 222. In some embodiments, the reflection sidewall thickness 232 is greater than or equal to three times the adhesion sidewall thickness 222. In some embodiments, a top surface of the electromagnetic reflection structure 234 and a top surface of the adhesion structure 236 are separated by the top side 108 of the semiconductor substrate 102 by a top side distance 238. The top surface of the electromagnetic reflection structure 234 and the top surface of the adhesion structure 236 extend into the semiconductor substrate 102 by the top side distance 238 from the top side 108 of the semiconductor substrate 102.

In some embodiments, portions of the adhesion layer 214 and portions of the reflection layer 224 above the top side 108 of the semiconductor substrate 102 are removed to retain dielectric layer portions within the recesses 130, such as the first dielectric layer portion 148 within the recess 132. In some embodiments, the portions of the adhesion layer 214 and the portions of the reflection layer 224 above the top side 108 of the semiconductor substrate 102 are removed in a same processing operation. The portions of the adhesion layer 214 and the portions of the reflection layer 224 are removed by at least one of chemical-mechanical polishing (CMP), etching, or other suitable techniques. The etching process is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material.

In some embodiments, the reflection structures are at least one of horizontally coincident with one or more of the semiconductor devices 112, horizontally coincident with a portion of one or more of the semiconductor devices 112, horizontally offset from one or more of the semiconductor devices 112, or horizontally offset from a portion of one or more of the semiconductor devices 112. For example, as illustrated in FIG. 1P, the electromagnetic reflection structure 234 comprising the first reflection sidewall 226, the second reflection sidewall 228, and the third reflection sidewall 230 is horizontally offset from a portion of the first

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semiconductor device **112a** and horizontally offset from a portion of the second semiconductor device **112b**.

In some embodiments, the electromagnetic reflection structure **234** comprises at least one of the first reflection sidewall **226** or the second reflection sidewall **228**. The first reflection sidewall **226** has a first reflection slope, such as a negative slope, or the second reflection sidewall **228** has a second reflection slope, such as a positive slope. In some embodiments, the second reflection slope is opposite in polarity relative to the first reflection slope. In some embodiments, the electromagnetic reflection structure **234** has a trapezoidal shape such that the third reflection sidewall **230** connects the first reflection sidewall **226** to the second reflection sidewall **228**. In some embodiments, the third reflection sidewall **230** is perpendicular to the direction **113**. In some embodiments, a cross-sectional area of the electromagnetic reflection structure **234** decreases along the direction **113**, such that a width of an upper portion of the electromagnetic reflection structure **234** is greater than a width of a lower portion of the electromagnetic reflection structure **234**.

In some embodiments, the adhesion structures are at least one of horizontally coincident with one or more of the semiconductor devices **112**, horizontally coincident with a portion of one or more of the semiconductor devices **112**, horizontally offset from one or more of the semiconductor devices **112**, or horizontally offset from a portion of one or more of the semiconductor devices **112**. For example, as illustrated in FIG. 1P, the adhesion structure **236** comprising the first adhesion sidewall **216**, the second adhesion sidewall **218**, and the third adhesion sidewall **220** is horizontally offset from a portion of the first semiconductor device **112a** and horizontally offset from a portion of the second semiconductor device **112b**.

In some embodiments, the adhesion structure **236** comprises at least one of the first adhesion sidewall **216** or the second adhesion sidewall **218**. The first adhesion sidewall **216** has a first adhesion slope, such as a negative slope, or the second adhesion sidewall **218** has a second adhesion slope, such as a positive slope. In some embodiments, the second adhesion slope is opposite in polarity relative to the first adhesion slope. In some embodiments, the adhesion structure **236** has a trapezoidal shape such that the third adhesion sidewall **220** connects the first adhesion sidewall **216** to the second adhesion sidewall **218**. In some embodiments, the third adhesion sidewall **220** is perpendicular to the direction **113**. In some embodiments, a cross-sectional area of the adhesion structure **236** decreases along the direction **113**, such that a width of an upper portion of the adhesion structure **236** is greater than a width of a lower portion of the adhesion structure **236**. Other configurations for removal of the portions of the adhesion layer **214** and the portions of the reflection layer **224** are within the scope of the present disclosure.

FIG. 1Q illustrates a second buffer layer **240** formed over the semiconductor substrate **102**, over a top surface of the buffer structure **212**, over a top surface of the adhesion structure **236**, over a top surface of the electromagnetic reflection structure **234**, and in the shallow trenches **166**, according to some embodiments. In some embodiments, the second buffer layer **240** is in direct contact with at least one of the top side **108** of the semiconductor substrate **102**, one or more of the first dielectric layer portions (e.g., the first dielectric layer portion **148**), the top surface of the buffer structure **212**, the top surface of the adhesion structure **236**, or the top surface of the electromagnetic reflection structure **234**. In some embodiments, the second buffer layer **240** is in

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indirect contact with at least one of the top side **108** of the semiconductor substrate **102**, one or more of the first dielectric layer portions (e.g., the first dielectric layer portion **148**), the top surface of the buffer structure **212**, the top surface of the adhesion structure **236**, or the top surface of the electromagnetic reflection structure **234**. In some embodiments, the second buffer layer **240** is formed as a single layer in a single manufacturing operation. In some embodiments, the second buffer layer **240** is formed as multiple layers of the same material in multiple manufacturing operations.

In some embodiments, the second buffer layer **240** comprises at least one of SiO, SiO₂, HfSiON, HfSiO_x, HfAlO_x, HfO₂, ZrO₂, La₂O₃, Y₂O₃, silicon nitride (Si₃N₄), silicon oxynitride (SiO_xN_y), hafnium oxide, spin-on glass (SOG), fluorinated silica glass (FSG), carbon doped silicon oxide (e.g., SiCOH), a high-density-plasma (HDP) oxide, an HDP-CVD oxide, or other suitable material. In some embodiments, the second buffer layer **240** comprises the same material as the buffer layer **202**. In some embodiments, the buffer layer **202** is a first oxide layer and the second buffer layer **240** is a second oxide layer, where the first buffer layer **202** has a greater refractive index than the second buffer layer **240**. In some embodiments, the second buffer layer **240** is a second oxide layer that has a second oxide top layer thickness greater than or equal to 100 Å over the top surface of the adhesion structure **236** and the top surface of the electromagnetic reflection structure **234**. The second buffer layer **240** is formed by at least one of PVD, sputtering, CVD, HDP-CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, portions of the second buffer layer **240** at least one of fill each of the shallow trenches **166** or partially fill each of the shallow trenches **166**, such as the shallow trench **168**. Other structures and/or configurations of the second buffer layer **240** are within the scope of the present disclosure.

FIG. 1R illustrates removal of portions of the second buffer layer **240**, according to some embodiments. In some embodiments, portions of the second buffer layer **240** above the top side **108** of the semiconductor substrate **102** are removed to retain fill structures, such as a fill structure **242**. In some embodiments, portions of the second buffer layer **240** above the top side **108** of the semiconductor substrate **102** are removed to retain dielectric layer portions within the recesses **130**, such as the first dielectric layer portion **148** within the recess **132**. The portions of the second buffer layer **240** are removed by at least one of chemical-mechanical polishing (CMP), etching, or other suitable techniques. The etching process is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material.

In some embodiments, the fill structures, such as the fill structure **242**, are at least one of horizontally coincident with one or more of the semiconductor devices **112**, horizontally coincident with a portion of one or more of the semiconductor devices **112**, horizontally offset from one or more of the semiconductor devices **112**, or horizontally offset from a portion of one or more of the semiconductor devices **112**. For example, as illustrated in FIG. 1R, the fill structure **242** is horizontally offset from a portion of the first semiconductor device **112a** and horizontally offset from a portion of the second semiconductor device **112b**. As set forth herein and in some embodiments, one or more of the semiconductor devices **112** or portions of the semiconductor devices **112** may be coincident with the tapered sidewalls of the shallow

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trenches 166 such that portions of the semiconductor devices 112 are horizontally offset from the fill structure 242 that aligns with the tapered sidewalls.

In some embodiments, the fill structure 242 comprises at least one of a first fill side 244 or a second fill side 246. The first fill side 244 may align with the first reflection sidewall 226 and the second fill side 246 may align with the second reflection sidewall 228. The first fill side 244 has a first fill slope, such as a negative slope, or the second fill side 246 has a second fill slope, such as a positive slope. In some embodiments, the second fill slope is opposite in polarity relative to the first fill slope. In some embodiments, the fill structure 242 has a portion forming a trapezoidal shape such that a third fill side 248 connects the first fill side 244 to the second fill side 246. In some embodiments, the third fill side 248 is perpendicular to the direction 113. In some embodiments, a cross-sectional area of the fill structure 242 decreases along the direction 113, such that a width of an upper portion of the fill structure 242 is greater than a width of a lower portion of the fill structure 242.

In some embodiments, the fill structure 242 has a fourth fill side 250, a fifth fill side 252, a sixth fill side 254, a seventh fill side 256, and an eighth fill side 258. The fourth fill side 250 connects the first fill side 244 with the fifth fill side 252, the fifth fill side 252 connects the fourth fill side 250 with the sixth fill side 254, the sixth fill side 254 connects the fifth fill side 250 with the seventh fill side 256, and the seventh fill side 256 connects the sixth fill side 254 with the second fill side 246. In some embodiments, the fourth fill side 250 or the eighth fill side 258 are perpendicular to the direction 113. In some embodiments, the buffer structure 212 comprises a first oxide layer formed within the shallow trench 168 of the semiconductor substrate 102 and the fill structure 242 comprises a second oxide layer that covers a top surface of the adhesion structure 236, a top surface of the electromagnetic reflection structure 234, and an inner surface of the electromagnetic reflection structure 234. Other processes and/or techniques for forming the fill structure 242 are within the scope of the present disclosure.

FIG. 1S illustrates the fill structures encapsulating electromagnetic reflection structures and adhesion structures, according to some embodiments. In some embodiments, the fill structure 242 and the buffer structure 212, as illustrated in FIG. 1R, comprise the same type material and comprise a second fill structure 260. In some embodiments, the second fill structure 260 completely surrounds the electromagnetic reflection structure 234 and the adhesion structure 236. In some embodiments, a top surface of the electromagnetic reflection structure 234 and a top surface of the adhesion structure 236 are separated by the top side 108 of the semiconductor substrate 102 by the top side distance 238 of the second fill structure 260. The top surface of the electromagnetic reflection structure 234 and the top surface of the adhesion structure 236 extend into the semiconductor substrate 102 by the top side distance 238 from the top side 108 of the semiconductor substrate 102 while being surrounded by the second fill structure 260.

In some embodiments, the semiconductor arrangement 100 includes one or more semiconductor structures, such as a semiconductor structure 268. The semiconductor structure 268 includes one or more shallow trench isolation structures 270 (STI structures), such as a first STI structure 272 and a second STI structure 274. The first STI structure 272 is within the semiconductor substrate 102. The first STI structure 272 includes the buffer structure 212, the adhesion structure 236, the electromagnetic reflection structure 234, and the fill structure 242. The adhesion structure 236 is

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between and adhesively bonded to the buffer structure 212 and the electromagnetic reflection structure 234. The electromagnetic reflection structure 234 is between the adhesion structure 236 and the fill structure 242 to reflect electromagnetic radiation. In some embodiments, the first STI structure 272 is laterally offset from a semiconductor device, such as the first semiconductor device 112a. In some embodiments, the second STI structure 274 is laterally offset from a semiconductor device, such as the second semiconductor device 112b, such that the second semiconductor device 112b is between the first STI structure 272 and the second STI structure 274. In some embodiments, the first STI structure 272 extends into the semiconductor substrate 102 a first depth (e.g., the top side distance 238) from the top side 108 of the semiconductor substrate 102 and the second semiconductor device 112b extends into the semiconductor substrate 102 a second depth (e.g., the device distance 138), less than the first depth, from the top side 108 of the semiconductor substrate 102. Other configurations and/or arrangements of the second fill structure 260 are within the scope of the present disclosure.

FIG. 1T illustrates a second dielectric layer 280 formed over the semiconductor substrate 102, according to some embodiments. In some embodiments, the second dielectric layer 280 is in direct contact with at least one of the top side 108 of the semiconductor substrate 102, one or more of the first dielectric layer portions (e.g., the first dielectric layer portion 148), or the top sides of the second fill structures (e.g., the sixth fill side 254 of the second fill structure 260). In some embodiments, the second dielectric layer 280 is in indirect contact with at least one of the top side 108 of the semiconductor substrate 102, one or more of the first dielectric layer portions (e.g., the first dielectric layer portion 148), or the top sides of the second fill structures (e.g., the sixth fill side 254 of the second fill structure 260).

The second dielectric layer 280 comprises at least one of SiO, SiO₂, SiN, Si₃N₄, MgO, Al₂O₃, Yb₂O₃, ZnO, Ta₂O₅, ZrO₂, HfO₂, TeO₂, TiO₂, an oxide layer, or other suitable material. The first dielectric layer 144 is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, the second dielectric layer 280 comprises the same material type as the first dielectric layer 144.

In some embodiments, the semiconductor arrangement 100 comprises a buffer layer (not shown) between the semiconductor substrate 102 and the second dielectric layer 280, as described in greater detail here. In some embodiments, the buffer layer is in direct contact with at least one of the top side 108 of the semiconductor substrate 102, one or more of the first dielectric layer portions (e.g., the first dielectric layer portion 148), or the top sides of the second fill structures (e.g., the sixth fill side 254 of the second fill structure 260). In some embodiments, the buffer layer is in indirect contact with at least one of the top side 108 of the semiconductor substrate 102, one or more of the first dielectric layer portions (e.g., the first dielectric layer portion 148), or the top sides of the second fill structures (e.g., the sixth fill side 254 of the second fill structure 260). Other structures and/or configurations of the second dielectric layer 280 are within the scope of the present disclosure.

FIG. 2 is a cross-sectional view of a semiconductor arrangement 100, according to some embodiments. In some embodiments, the semiconductor structure 268 is formed without the dielectric layer portions, such as the first dielectric layer portion 148, as illustrated in FIG. 1M. Hence, the recesses 130 as illustrated in FIG. 1F are not formed in the

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semiconductor substrate **102**. As illustrated in FIG. 2, the top side **108** of the semiconductor substrate **102** is generally perpendicular to the direction **113**. In some embodiments, the second dielectric layer **280** is in direct contact with at least one of the top side **108** of the semiconductor substrate **102**, or the top sides of the second fill structures (e.g., the sixth fill side **254** of the second fill structure **260**). In some embodiments, the second dielectric layer **280** is in indirect contact with at least one of the top side **108** of the semiconductor substrate **102**, or the top sides of the second fill structures (e.g., the sixth fill side **254** of the second fill structure **260**). Other structures and/or configurations of the semiconductor structure **268** are within the scope of the present disclosure.

FIG. 3 is a cross-sectional view of a semiconductor arrangement **100**, according to some embodiments. In some embodiments, one or more of the semiconductor devices, such as the second semiconductor device **112b**, are arranged as a photodiode **300** formed over the semiconductor substrate **102**. The photodiode **300** comprises a charge storage well **308**, a pinning layer **310**, a transfer gate **312**, and a floating diffusion **314**. In some embodiments, the photodiode **300** comprises a first deep storage well section **316** and a second deep storage well section **318**. In FIG. 3, the pinning layer **310** is arranged between the second region **102b** and the charge storage well **308**. In some embodiments, elements of the photodiode **300** are adjacent to an STI structure, such as the first STI structure **272** or the second STI structure **274**. In some embodiments, elements of the photodiode **300** are directly adjacent to an STI structure, such as the first STI structure **272** or the second STI structure **274**. In some embodiments, the STI structures, such as the first STI structure **272**, are at least one of horizontally coincident with the photodiode **300**, horizontally coincident with a portion of the photodiode **300**, horizontally offset from the photodiode **300**, or horizontally offset from a portion of the photodiode **300**. For example, as illustrated in FIG. 3, the first STI structure **272** is horizontally offset from the photodiode **300** and the second STI structure **274** is horizontally offset from the photodiode **300**.

According to some embodiments, a ratio **302** of thickness of the adhesion layer **236** to the reflection layer **234** (e.g., Ti/TiN thickness) is greater than ($>$) 1:3. The ratio **302** is not merely a matter of design choice, but is critical to, at least, provide sufficient reflectivity of the shallow trench **168** while not consuming too much space. As set forth in greater detail herein and in some embodiments, the adhesion layer **236** and the reflection layer **234** (e.g., a Ti/TiN layer) are fully buried inside the second fill structure **260** (e.g., an oxide film) to mitigate against contamination. As set forth in greater detail herein and in some embodiments, a distance **304** from the adhesion layer **236** and the reflection layer **234** (e.g., a Ti/TiN layer) to the oxide surface (e.g., below the second dielectric layer **280**) is greater than ($>$) 100 Å.

In some embodiments, an ARC layer **320** (i.e., an anti-reflective coating layer) is arranged over the photodiode **300** and over the semiconductor substrate **102**. A radiation filter layer **322** is arranged over the ARC layer **320**, over the photodiode **300**, and over the semiconductor substrate **102**. The radiation filter layer **322** comprises a plurality of radiation pass filters **324**, each configured to allow a radiation pass range of wavelengths to pass through and be detected by the corresponding photodiode. In some embodiments, a micro-lens array **326** is arranged over the ARC layer **320** and the radiation filter layer **322**, according to some embodiments. The micro-lens array **326** is arranged to steer radiation towards underlying photodiodes, such as the

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photodiode **300**. The micro-lens array **326** includes a plurality of micro-lenses configured to transmit radiation to a corresponding photodiode.

FIG. 4 illustrates an example method **400** for forming a semiconductor structure according to some embodiments. Some of the operations described can be replaced and/or eliminated for different embodiments. At **402**, a buffer structure comprising a first oxide layer is formed in a shallow trench of a semiconductor substrate. For example, in FIG. 1M, the buffer structure **212** is formed from the buffer layer **202** in the shallow trench **168** of the semiconductor substrate **102**, where the buffer layer **202** may comprise an oxide layer.

At **404**, an adhesion structure comprising titanium is formed over the buffer structure, wherein the adhesion structure has an adhesion sidewall thickness. For example, in FIG. 1P, the adhesion structure **236** is formed from the adhesion layer **214** over the buffer structure **212**, where the adhesion layer **214** may comprise titanium and the adhesion structure **236** may have the adhesion sidewall thickness **222**.

At **406**, an electromagnetic reflection structure comprising titanium nitride is formed over the adhesion structure, wherein the electromagnetic reflection structure has a reflection sidewall thickness greater than or equal to three times the adhesion sidewall thickness. For example, in FIG. 1P, the electromagnetic reflection structure **234** is formed over the adhesion structure **236**, where the electromagnetic reflection structure **234** may comprise titanium nitride and have the reflection sidewall thickness **232** greater than or equal to three times the adhesion sidewall thickness **222**.

At **408**, a fill structure comprising a second oxide layer is formed that covers a top surface of the adhesion structure, a top surface of the electromagnetic reflection structure, and an inner surface of the electromagnetic reflection structure. For example, in FIG. 1R, the fill structure **242** is formed from the second buffer layer **240** and covers the top surface of the adhesion structure **236**, the top surface of the electromagnetic reflection structure **234**, and the inner surface of the electromagnetic reflection structure **234**. The second buffer layer **240** may comprise a second oxide layer.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a first shallow trench isolation (STI) structure within a semiconductor substrate and the first STI structure comprises a buffer structure, an adhesion structure, an electromagnetic reflection structure, and a fill structure. The adhesion structure is between and adhesively bonded to the buffer structure and the electromagnetic reflection structure. The electromagnetic reflection structure is between the adhesion structure and the fill structure to reflect electromagnetic radiation.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a photodiode within a semiconductor substrate and a first shallow trench isolation (STI) structure laterally offset from the photodiode within the semiconductor substrate. The first STI structure includes a buffer structure comprising a first oxide layer and formed within a shallow trench of the semiconductor substrate and an adhesion structure comprising titanium and formed as an adhesion layer over the first oxide layer. An electromagnetic reflection structure comprising titanium nitride is formed as a reflection layer over the adhesion layer and a fill structure, comprising a second oxide layer, covers a top surface of the adhesion structure, a top surface of the electromagnetic reflection structure, and an inner surface of the electromagnetic reflection structure.

A method for forming a semiconductor structure is provided. The method includes forming a buffer structure

comprising a first oxide layer in a shallow trench of a semiconductor substrate and forming an adhesion structure comprising titanium over the buffer structure, wherein the adhesion structure has an adhesion sidewall thickness. The method includes forming an electromagnetic reflection structure comprising titanium nitride over the adhesion structure, wherein the electromagnetic reflection structure has a reflection sidewall thickness greater than or equal to three times the adhesion sidewall thickness. The method includes forming a fill structure comprising a second oxide layer that covers a top surface of the adhesion structure, a top surface of the electromagnetic reflection structure, and an inner surface of the electromagnetic reflection structure.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Although the subject matter has been described in language specific to structural features or methodological acts, it is to be understood that the subject matter of the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing at least some of the claims.

Various operations of embodiments are provided herein. The order in which some or all of the operations are described should not be construed to imply that these operations are necessarily order dependent. Alternative ordering will be appreciated having the benefit of this description. Further, it will be understood that not all operations are necessarily present in each embodiment provided herein. Also, it will be understood that not all operations are necessary in some embodiments.

It will be appreciated that layers, features, elements, etc. depicted herein are illustrated with particular dimensions relative to one another, such as structural dimensions or orientations, for example, for purposes of simplicity and ease of understanding and that actual dimensions of the same differ substantially from that illustrated herein, in some embodiments. Additionally, a variety of techniques exist for forming the layers, regions, features, elements, etc. mentioned herein, such as at least one of etching techniques, planarization techniques, implanting techniques, doping techniques, spin-on techniques, sputtering techniques, growth techniques, or deposition techniques such as CVD, for example.

Moreover, "exemplary" is used herein to mean serving as an example, instance, illustration, etc., and not necessarily as advantageous. As used in this application, "or" is intended to mean an inclusive "or" rather than an exclusive "or". In addition, "a" and "an" as used in this application and the appended claims are generally to be construed to mean "one or more" unless specified otherwise or clear from context to be directed to a singular form. Also, at least one of A and B and/or the like generally means A or B or both A and B. Furthermore, to the extent that "includes", "having", "has", "with", or variants thereof are used, such terms are intended to be inclusive in a manner similar to the term "comprising".

Also, unless specified otherwise, "first," "second," or the like are not intended to imply a temporal aspect, a spatial aspect, an ordering, etc. Rather, such terms are merely used as identifiers, names, etc. for features, elements, items, etc. For example, a first element and a second element generally correspond to element A and element B or two different or two identical elements or the same element.

Also, although the disclosure has been shown and described with respect to one or more implementations, equivalent alterations and modifications will occur to others of ordinary skill in the art based upon a reading and understanding of this specification and the annexed drawings. The disclosure comprises all such modifications and alterations and is limited only by the scope of the following claims. In particular regard to the various functions performed by the above described components (e.g., elements, resources, etc.), the terms used to describe such components are intended to correspond, unless otherwise indicated, to any component which performs the specified function of the described component (e.g., that is functionally equivalent), even though not structurally equivalent to the disclosed structure. In addition, while a particular feature of the disclosure may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

What is claimed is:

1. A semiconductor structure, comprising:

a first shallow trench isolation (STI) structure within a semiconductor substrate, wherein:

the first STI structure comprises a buffer structure, an adhesion structure, an electromagnetic reflection structure, and a fill structure;

the adhesion structure is between and adhesively bonded to the buffer structure and the electromagnetic reflection structure;

the electromagnetic reflection structure is between the adhesion structure and the fill structure to reflect electromagnetic radiation;

the buffer structure comprises a first oxide layer formed within a shallow trench of the semiconductor substrate; and

the fill structure comprises a second oxide layer that covers a top surface of the adhesion structure, a top surface of the electromagnetic reflection structure, and an inner surface of the electromagnetic reflection structure.

2. The semiconductor structure of claim 1, wherein the buffer structure is contiguous with a doped region of the semiconductor substrate; and

the buffer structure has a buffer sidewall thickness between about 100 Angstroms to about 500 Angstroms to buffer against dopants of the doped region.

3. The semiconductor structure of claim 1, wherein: the adhesion structure comprises titanium and is formed as an adhesion layer contiguous with the buffer structure.

4. The semiconductor structure of claim 1, wherein: the adhesion structure has an adhesion sidewall thickness greater than about 100 Angstroms.

5. The semiconductor structure of claim 1, wherein: the electromagnetic reflection structure comprises titanium nitride and is formed as a reflection layer contiguous with the adhesion structure.

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6. The semiconductor structure of claim 1, wherein:
the adhesion structure has an adhesion sidewall thickness;
and
the electromagnetic reflection structure has a reflection
sidewall thickness greater than the adhesion sidewall
thickness. 5
7. The semiconductor structure of claim 6, wherein:
the reflection sidewall thickness is greater than or equal to
three times the adhesion sidewall thickness. 10
8. The semiconductor structure of claim 1, wherein:
the second oxide layer has a second oxide top layer
thickness greater than or equal to 100 Angstroms over
the top surface of the adhesion structure and the top
surface of the electromagnetic reflection structure. 15
9. The semiconductor structure of claim 1, wherein the
first oxide layer has a greater refractive index than the
second oxide layer.
10. The semiconductor structure of claim 1, wherein:
the fill structure comprises at least one of silicon dioxide,
silicon oxynitride, or hafnium oxide. 20
11. The semiconductor structure of claim 1, wherein:
the buffer structure comprises at least one of aluminum
oxide or hafnium oxide. 25
12. The semiconductor structure of claim 1, wherein at
least one of:
the adhesion structure comprises a mixture of titanium
and a first metal; or
the electromagnetic reflection structure comprises a mix-
ture of titanium nitride and a second metal. 30
13. The semiconductor structure of claim 1, comprising:
a semiconductor device within the semiconductor sub-
strate, wherein the first STI structure is laterally offset
from the semiconductor device. 35
14. The semiconductor structure of claim 13, comprising:
a second STI structure within the semiconductor substrate
and laterally offset from the semiconductor device,
wherein: 40
- the semiconductor device is between the first STI struc-
ture and the second STI structure.
15. The semiconductor structure of claim 13, wherein:
the first STI structure extends into the semiconductor
substrate a first depth from a top side of the semicon-
ductor substrate; and 45
- the semiconductor device extends into the semiconductor
substrate a second depth, less than the first depth, from
the top side of the semiconductor substrate.

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16. A semiconductor structure, comprising:
a photodiode within a semiconductor substrate; and
a first shallow trench isolation (STI) structure laterally
offset from the photodiode within the semiconductor
substrate, comprising:
a buffer structure comprising a first oxide layer and
formed within a shallow trench of the semiconductor
substrate;
an adhesion structure comprising titanium and formed
as an adhesion layer over the first oxide layer;
an electromagnetic reflection structure comprising tita-
nium nitride and formed as a reflection layer over the
adhesion layer; and
a fill structure comprising a second oxide layer that
covers a top surface of the adhesion structure, a top
surface of the electromagnetic reflection structure,
and an inner surface of the electromagnetic reflection
structure.
17. The semiconductor structure of claim 16, wherein:
the shallow trench has a first trench tapered sidewall
tapered toward the photodiode;
the buffer structure has a first buffer tapered sidewall that
aligns with the first trench tapered sidewall;
the adhesion structure has a first adhesion tapered side-
wall that aligns with the first buffer tapered sidewall;
and
the electromagnetic reflection structure has a first reflec-
tion tapered sidewall that aligns with the first adhesion
tapered sidewall.
18. A semiconductor structure, comprising:
a first shallow trench isolation (STI) structure within a
semiconductor substrate, wherein:
the first STI structure comprises a buffer structure, an
adhesion structure, an electromagnetic reflection
structure, and a fill structure;
the adhesion structure is between and adhesively
bonded to the buffer structure and the electromag-
netic reflection structure;
the electromagnetic reflection structure is between the
adhesion structure and the fill structure to reflect
electromagnetic radiation; and
the adhesion structure comprises titanium and is
formed as an adhesion layer contiguous with the
buffer structure.
19. The semiconductor structure of claim 18, wherein:
the buffer structure comprises a first oxide layer; and
the fill structure comprises a second oxide layer.
20. The semiconductor structure of claim 19, wherein:
the second oxide layer covers a top surface of the elec-
tromagnetic reflection structure and an inner surface of
the electromagnetic reflection structure.

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